19 VECTOR DERIVATIVES

19.1 Directional derivatives

Suppose that $\boldsymbol{\xi}$ and $\mathbf{u}$ are vectors in $\mathbb{R}^n$ and that $\mathbf{u}$ is of unit length (i.e. $\|\mathbf{u}\| = 1$). We shall think of $\boldsymbol{\xi}$ as determining a point in $\mathbb{R}^n$ and of $\mathbf{u}$ as determining a direction. The set of all vectors of the form

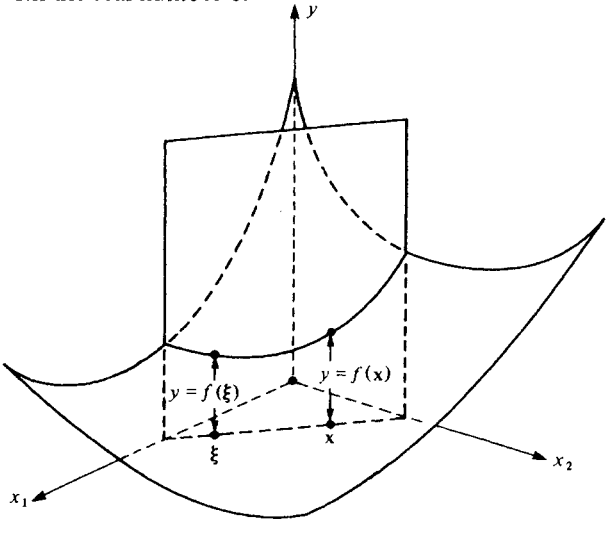
$$\mathbf{x} = \boldsymbol{\xi} + t\mathbf{u} \quad (t \in \mathbb{R})$$

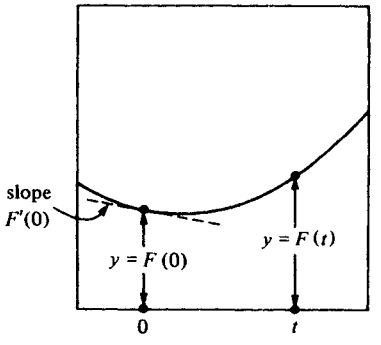
can then be thought of as the straight line through the point $\boldsymbol{\xi}$ in the direction of $\mathbf{u}$. The parameter t measures the distance from $\boldsymbol{\xi}$ to $\mathbf{x}$ along this line. (See §18.13.)

If $f\colon \mathbb{R}^n \to \mathbb{R}$, we can define a function $F\colon \mathbb{R} \to \mathbb{R}$ by

$$F(t) = f(\boldsymbol{\xi} + t\mathbf{u}).$$

In the case when $n = 2$, the geometry of the situation is very simple. One may regard $y = F(t)$ as the equation of the curve obtained by slicing the surface $y = f(\mathbf{x})$ with a vertical plane through the point $\boldsymbol{\xi}$ which is parallel to $\mathbf{u}$. The diagram below shows the surface $y = f(\mathbf{x})$ sliced by the appropriate vertical plane. The following diagram suppresses everything outside this vertical plane and relabels the point $\mathbf{x} = \boldsymbol{\xi} + t\mathbf{u}$ with the real number t. Correspondingly, $\boldsymbol{\xi}$ is relabelled with the real number 0.





It is apparent from these diagrams that $F'(0)$ is the slope (or rate of increase) of the surface $y = f(\mathbf{x})$ at the point $\boldsymbol{\xi}$ in the direction $\mathbf{u}$. We call this important quantity the *directional derivative* of f at the point $\boldsymbol{\xi}$ in the direction $\mathbf{u}$ and denote it by $D_{\mathbf{u}}f(\boldsymbol{\xi})$. Recalling the definition of a derivative given in §10.1, we have that

$$D_{\mathbf{u}}f(\boldsymbol{\xi}) = \lim_{t\to 0} \frac{f(\boldsymbol{\xi} + t\mathbf{u}) - f(\boldsymbol{\xi})}{t}.$$

The same notation is also used when f is a vector-valued function and $\mathbf{u}$ is not necessarily a unit vector. But then of course the preceding geometric interpretation does not apply.

19.2 Example Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(x_1, x_2) = x_1x_2$. Find the slope of the surface $y = f(x_1, x_2)$ at the point $(1, 2)$ in the direction $(3, 4)$.

The vector $(3, 4)$ is not a unit vector and so we begin by replacing it by the unit vector $(3/5, 4/5)$ which points in the same direction. We have to calculate

$$D_{\mathbf{u}}f(\boldsymbol{\xi}) = \lim_{t\to 0} \frac{f(\boldsymbol{\xi} + t\mathbf{u}) - f(\boldsymbol{\xi})}{t}$$

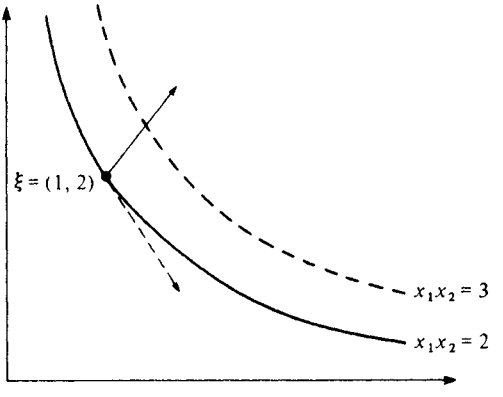
in the case when $\boldsymbol{\xi} = (1, 2)$ and $\mathbf{u} = (3/5, 4/5)$. We obtain that

$$D_{\mathbf{u}}f(\boldsymbol{\xi}) = \lim_{t\to 0} \frac{(1 + 3t/5)(2 + 4t/5) - (1)(2)}{t}$$

$$= \lim_{t\to 0} \left(2 + \frac{12t}{25}\right) = 2.$$

It follows that the slope of the surface $y = x_1x_2$ at the point $(1, 2)$ in the direction $(3, 4)$ is equal to 2. The diagram illustrates the curve $x_1x_2 = 2$ which is called a *contour* of the function f. Points on the surface $y = f(x_1, x_2)$ above this contour are all at the same height. The unbroken arrow drawn at $(1, 2)$ indicates

the direction (3, 4). The broken arrow indicates the direction $(1, -2)$ which is tangent to the contour $x_1x_2 = 2$ (why?). It is instructive to check that the directional derivative at (1, 2) in the direction $(1, -2)$ is zero.



19.3 Partial derivatives

Suppose that f: $\mathbb{R}^n \to \mathbb{R}$. To avoid cumbersome notation we shall assume in what follows that $n = 2$ but the general case is entirely analogous.

If $\boldsymbol{\xi} \in \mathbb{R}^2$ the *partial derivatives* of f at $\boldsymbol{\xi}$ are defined by

$$f_1(\xi_1, \xi_2) = \lim_{h \to 0} \frac{f(\xi_1 + h, \xi_2) - f(\xi_1, \xi_2)}{h}$$

$$f_2(\xi_1, \xi_2) = \lim_{k \to 0} \frac{f(\xi_1, \xi_2 + k) - f(\xi_1, \xi_2)}{k}$$

provided that these limits exist.

It is no more difficult to calculate the partial derivatives of a function of a vector variable than it is to calculate the derivative of a function of a real variable. To calculate f_1, for example, one simply differentiates with respect to x_1 treating x_2 as a constant.

19.4 Examples

(i) Let f: $\mathbb{R}^2 \to \mathbb{R}$ be defined by $f(x_1, x_2) = x_1{}^2x_2 + x_2{}^3x_1$. Then

$$\left.\begin{aligned} f_1(x_1, x_2) &= 2x_1x_2 + x_2{}^3 \\ f_2(x_1, x_2) &= x_1{}^2 + 3x_2{}^2x_1. \end{aligned}\right\}$$

(ii) Let g: $\mathbb{R}^2 \to \mathbb{R}$ be defined by $g(x_1, x_2) = \sin(x_1x_2 + x_2^2)$. Then

$$\left.\begin{aligned} g_1(x_1, x_2) &= x_2 \cos(x_1x_2 + x_2^2) \\ g_2(x_1, x_2) &= (x_1 + 2x_2)\cos(x_1x_2 + x_2^2). \end{aligned}\right\}$$

(iii) Let h: $\mathbb{R}^3 \to \mathbb{R}$ be defined by $h(x_1, x_2, x_3) = x_1 + x_2x_3$. Then

$$\left.\begin{aligned} h_1(x_1, x_2, x_3) &= 1 \\ h_2(x_1, x_2, x_3) &= x_3 \\ h_3(x_1, x_2, x_3) &= x_2. \end{aligned}\right\}$$

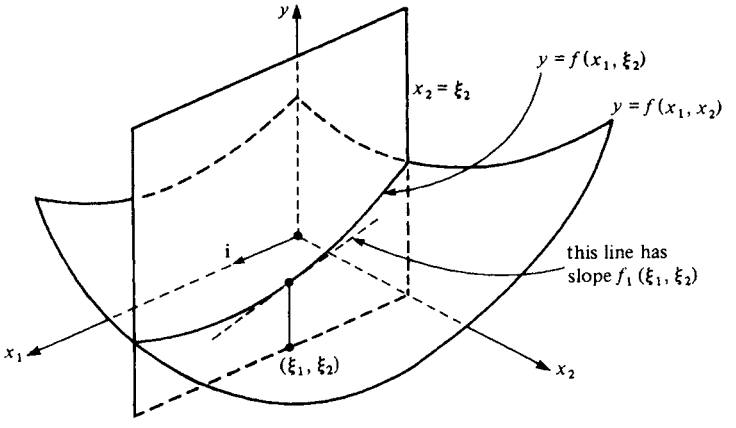
The geometric interpretation of a partial derivative is important. Let $\mathbf{i} = (1, 0)$ and $\mathbf{j} = (0, 1)$. Observe that $\mathbf{i}$ is a unit vector which points in the direction of the x_1-axis while $\mathbf{j}$ is a unit vector which points in the direction of the x_2-axis. We have that

$$\begin{aligned} D_{\mathbf{i}}f(\boldsymbol{\xi}) &= \lim_{t\to 0} \frac{f(\boldsymbol{\xi} + t\mathbf{i}) - f(\boldsymbol{\xi})}{t} \\ &= \lim_{t\to 0} \frac{f(\xi_1 + t, \xi_2) - f(\xi_1, \xi_2)}{t} \\ &= f_1(\xi_1, \xi_2). \end{aligned}$$

Similarly,

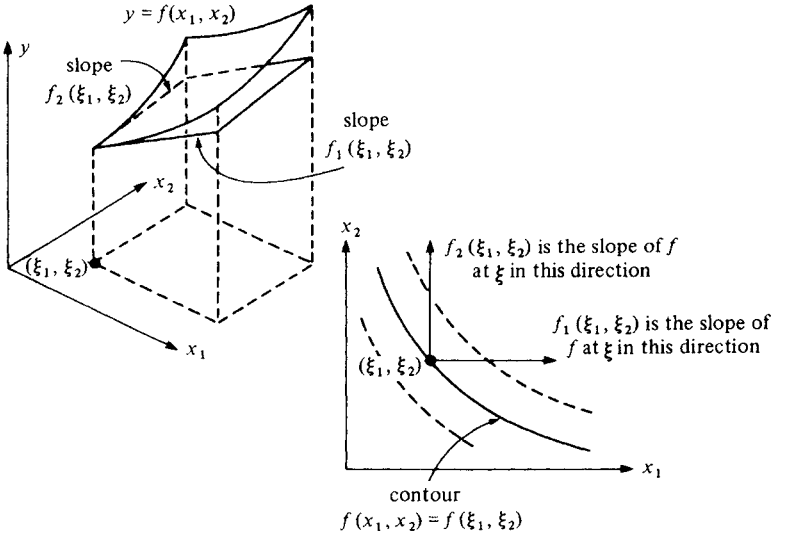
$$\begin{aligned} D_{\mathbf{j}}f(\boldsymbol{\xi}) &= \lim_{t\to 0} \frac{f(\boldsymbol{\xi} + t\mathbf{j}) - f(\boldsymbol{\xi})}{t} \\ &= \lim_{t\to 0} \frac{f(\xi_1, \xi_2 + t) - f(\xi_1, \xi_2)}{t} \\ &= f_2(\xi_1, \xi_2). \end{aligned}$$

Thus the partial derivative $f_1(\xi_1, \xi_2)$ is the slope of the surface $y = f(x_1, x_2)$ at the point $\boldsymbol{\xi}$ in the direction of the x_1-axis while $f_2(\xi_1, \xi_2)$ is the slope in the direction of the x_2-axis.



The diagram above illustrates the geometric significance of the partial derivative $f_1(\xi_1, \xi_2)$. It shows the surface $y = f(x_1, x_2)$ sliced by the vertical plane $x_2 = \xi_2$ (which passes through (ξ_1, ξ_2) and is parallel to $\mathbf{i} = (1, 0)$). On the plane $x_2 = \xi_2$, the graph $y = f(x_1, \xi_2)$ can be seen. The partial derivative $f_1(\xi_1, \xi_2)$ is simply the slope of this graph at the point $x_1 = \xi_1$.

The diagrams below illustrate both partial derivatives simultaneously.



19.5 Notation

As in the case of ordinary derivatives, there are numerous alternative notations which have to be kept in mind when working with partial derivatives. Which is used will depend on the context.

We note first that it is common to denote the partial derivative $f_1(\xi_1, \xi_2)$ by

$$f_{x_1}(\xi_1, \xi_2)$$

with a similar convention for $f_2(\xi_1, \xi_2)$. This may seem a clumsy piece of notation but it is quite useful when, as often happens, it is convenient to write x rather than x_1 and y rather than x_2.

However, the most important alternative notation is the use of

$$\frac{\partial y}{\partial x_1}$$

for the partial derivative $f_1(x_1, x_2)$ in the case when $y = f(x_1, x_2)$. Similarly

$$\frac{\partial y}{\partial x_2}$$

is used for the partial derivative $f_2(x_1, x_2)$. As noted under similar circumstances in §10.4, this notation suffers from the disadvantage that x_1 and x_2 are used ambiguously both as the co-ordinates of the point at which the partial derivative is to be evaluated and as the variables with respect to which one is to differentiate. This ambiguity is often irrelevant in practical applications but can be very confusing when theorems are to be proved.

When a number of variables are in play, it is not always obvious which are to be held constant when differentiating partially. But it is of the *greatest* importance to be clear on this point. Where necessary, the notation

$$\left(\frac{\partial y}{\partial x_1}\right)_{x_2}$$

is available. This means 'differentiate partially with respect to x_1 treating x_2 as constant'.

19.6 *Example* Consider the equations

$$\left.\begin{aligned} u &= x + y \\ v &= x - y. \end{aligned}\right\}$$

From the first equation

$$\frac{\partial u}{\partial x} = 1$$

If the equations are solved for x and y in terms of u and v, we obtain that

$$\left.\begin{aligned} x &= \tfrac{1}{2}(u + v) \\ y &= \tfrac{1}{2}(u - v) \end{aligned}\right\}$$

and therefore that

$$\frac{\partial x}{\partial u} = \tfrac{1}{2}.$$

Note that

$$\frac{\partial u}{\partial x} \neq \left(\frac{\partial x}{\partial u}\right)^{-1}.$$

This is not in the least surprising since *different* variables were held constant during the two differentiations. What has been shown is that

$$\left(\frac{\partial u}{\partial x}\right)_y = 1 \quad \left(\frac{\partial x}{\partial u}\right)_v = \tfrac{1}{2}.$$

It is instructive to calculate

$$\left(\frac{\partial u}{\partial x}\right)_v.$$

Eliminating y from the first set of equations, we obtain that $u = 2x - v$ and hence

$$\left(\frac{\partial u}{\partial x}\right)_v = 2 = \left(\frac{\partial x}{\partial u}\right)_v^{-1}.$$

(Note: the symbol ∂ used in the notation for a partial derivative discussed above must be carefully distinguished from the Greek letter δ and the Roman letter d.)

19.7 Local maxima and minima

An immediate application of the ideas introduced above is to the location of local maxima and minima. These are defined as in §11.1. Suppose that $f: A \to \mathbb{R}$ where A contains every $\mathbf{x} \in \mathbb{R}^n$ satisfying $\|\mathbf{x} - \boldsymbol{\xi}\| < \Delta$. Then we say that f has a *local maximum* at the point $\boldsymbol{\xi}$ if and only if there exists a $\delta > 0$ such that

$$f(\mathbf{x}) \leqslant f(\boldsymbol{\xi})$$

for each $\mathbf{x}$ satisfying $\|\mathbf{x} - \boldsymbol{\xi}\| < \delta$. Similarly, f has a *local minimum* at $\boldsymbol{\xi}$ if and only if there exists a $\delta > 0$ such that

$$f(\mathbf{x}) \geqslant f(\boldsymbol{\xi})$$

for each $\mathbf{x}$ satisfying $\|\mathbf{x} - \boldsymbol{\xi}\| < \delta$.

Very roughly, if f has a local maximum at $\boldsymbol{\xi}$, then its graph has a little hill above $\boldsymbol{\xi}$ and, if f has a local minimum at $\boldsymbol{\xi}$, then its graph has a little valley above $\boldsymbol{\xi}$.

19.8 Theorem Suppose that $f: A \to \mathbb{R}$ where A contains every $\mathbf{x} \in \mathbb{R}^n$ satisfying $\|\mathbf{x} - \boldsymbol{\xi}\| < \Delta$ and that the directional derivative $D_{\mathbf{u}} f(\boldsymbol{\xi})$ exists. If f has a local maximum or minimum at $\boldsymbol{\xi}$, then

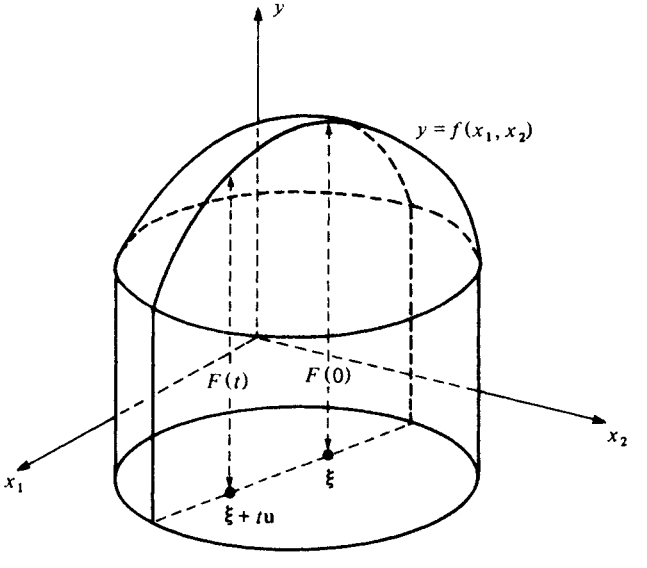
$$D_{\mathbf{u}} f(\boldsymbol{\xi}) = 0.$$

Proof A function F is defined on some open interval containing 0 by the formula

$$F(t) = f(\boldsymbol{\xi} + t\mathbf{u}).$$

If f has a local maximum or minimum at $\boldsymbol{\xi}$, then F has a local maximum or minimum at 0. From theorem 11.2, it follows that $F'(0) = 0$, i.e.

$$D_{\mathbf{u}} f(\boldsymbol{\xi}) = 0.$$



19.9 Corollary Suppose that $f: A \to \mathbb{R}^n$ where A contains every $\mathbf{x} \in \mathbb{R}^n$ satisfying $\|\mathbf{x} - \boldsymbol{\xi}\| < \Delta$. If f has a local maximum or minimum at $\boldsymbol{\xi}$, then all the partial derivatives of f which exist at $\boldsymbol{\xi}$ are zero

Proof As we know from § 19.3, a partial derivative is simply a directional derivative taken parallel to one of the co-ordinate axes.

19.10 Example Consider the function $f: \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x_1, x_2) = x_1^2 x_2 + x_2^3 x_1 - x_1 x_2.$$

By the preceding corollary, a necessary condition that f have a local maximum or minimum at (x_1, x_2) is that the equations

$$\left.\begin{aligned} 2x_1x_2 + x_2^3 - x_2 &= 0 \\ x_1^2 + 3x_2^2x_1 - x_1 &= 0 \end{aligned}\right\}$$

hold simultaneously. This means that

$$(x_2 = 0 \text{ or } 2x_1 + x_2^2 = 1) \text{ and } (x_1 = 0 \text{ or } x_1 + 3x_2^2 = 1)$$

i.e. $(x_2 = 0$ and $x_1 = 0)$ or $(x_2 = 0$ and $x_1 + 3x_2^2 = 1)$ or

$(2x_1 + x_2^2 = 1$ and $x_1 = 0)$ or $(2x_1 + x_2^2 = 1$ and $x_1 + 3x_2^2 = 1)$.

The only points at which it is possible for f to have a local maximum or minimum are therefore $(0, 0)$, $(1, 0)$, $(0, 1)$, $(0, -1)$, $(2/5, 1/\sqrt{5})$, $(2/5, -1/\sqrt{5})$. (The final pair of points is obtained by solving

$$\left.\begin{aligned} 2x_1 + x_2^2 &= 1 \\ x_1 + 3x_2^2 &= 1 \end{aligned}\right\}$$

simultaneously.)

It should be emphasised that all that has been shown is that f cannot have a local maximum or minimum at a point other than those listed. It has *not* been shown that f actually does have a local maximum or minimum at any of these points. In fact, f does *not* have a local maximum or minimum at $(0, 0)$. To see this, consider what happens on the lines $x_1 = x_2$ and $x_1 = -x_2$. We have that

$$f(t, t) = t^3 + t^4 - t^2$$

which has a local *maximum* at $t = 0$ while

$$f(t, -t) = t^3 - t^4 + t^2$$

which has a local *minimum* at $t = 0$. The point $(0, 0)$ is an example of a *saddle point* for f. We shall discuss such points in more detail in §19.40.

19.11 Exercise

(1) Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(x_1, x_2) = x_1^2x_2 + x_2^3x_1$. Compute the directional derivative of f at the point $(1, 2)$ in the direction $(3, 4)$.

(2) Compute the partial derivatives of the functions $f\colon \mathbb{R}^2 \to \mathbb{R}$ defined by

(i) $f(x_1, x_2) = x_1 + 2x_2$ (ii) $f(x_1, x_2) = x_1^2x_2^5$

(iii) $f(x_1, x_2) = \log(x_2 + e^{x_1x_2})^2$ (iv) $f(x_1, x_2) = \cos(x_2 \sin x_1)$.

(3) Find all the points in $\mathbb{R}^2$ at which all partial derivatives of the function $f\colon \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x_1, x_2) = x_1^4x_2 - x_1^2x_2^3$ are zero.

(4) Find all points in $\mathbb{R}^3$ at which all partial derivatives of the function $f\colon \mathbb{R}^3 \to \mathbb{R}$ defined by $f(x_1, x_2, x_3) = x_1x_2x_3$ are zero. Show that none of these points corresponds to a local maximum or minimum.

(5) The *component functions* f^i: $\mathbb{R}^n \to \mathbb{R}$ of a function f: $\mathbb{R}^n \to \mathbb{R}^m$ are defined by

$$f(\mathbf{x}) = (f^1(\mathbf{x}), f^2(\mathbf{x}), \ldots, f^m(\mathbf{x})).$$

Explain why

$$D_{\mathbf{u}}f(\boldsymbol{\xi}) = (D_{\mathbf{u}}f^1(\boldsymbol{\xi}), D_{\mathbf{u}}f^2(\boldsymbol{\xi}), \ldots, D_{\mathbf{u}}f^m(\boldsymbol{\xi})).$$

(6) Prove that the function f: $\mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x_1, x_2) = \begin{cases} \dfrac{x_1^2 x_2}{x_1^4 + x_2^2} & (x_1, x_2) \neq (0, 0) \\ 0 & (x_1, x_2) = (0, 0) \end{cases}$$

has directional derivatives in every direction at (0, 0) but is *not* continuous at (0, 0). [Hint: example 18.43.] Does f have a local maximum or minimum at (0, 0)?

19.12 Differentiable functions

Consider a function f: $A \to \mathbb{R}^m$ where A is a set which contains each $\mathbf{x} \in \mathbb{R}^n$ satisfying $\|\mathbf{x} - \boldsymbol{\xi}\| < \Delta$. What should it mean to say that f is differentiable at $\boldsymbol{\xi}$? We dealt with the case $m = n = 1$ in §10.1. Our definition required the existence of the limit

$$l = \lim_{x \to \xi} \frac{f(x) - f(\xi)}{x - \xi} = \lim_{h \to 0} \frac{f(\xi + h) - f(\xi)}{h}.$$

This may be rewritten as

$$f(x) - f(\xi) - l(x - \xi) = o(x - \xi) \qquad (x \to \xi)$$

or as

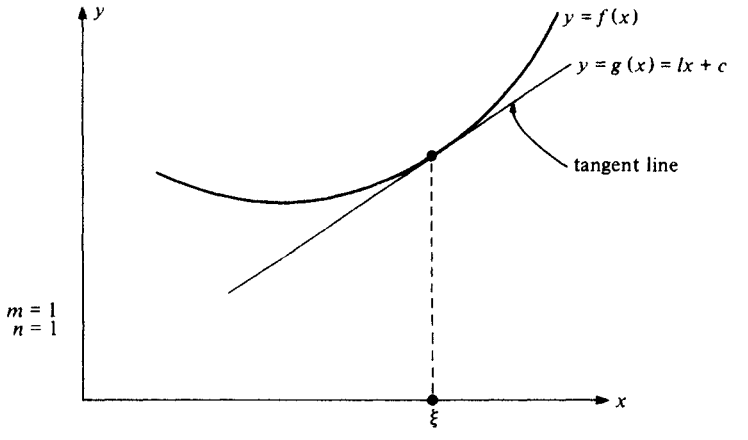
$$f(\xi + h) - f(\xi) - lh = o(h) \qquad (h \to 0)$$

where $o(h)$ denotes a quantity which tends to zero when divided by h. The number l, of course, is called the derivative of f at ξ and denoted by $f'(\xi)$.

An affine function g: $\mathbb{R} \to \mathbb{R}$ is defined by an equation of the form $g(x) = lx + c$. (See example 18.24(i).) We can therefore restate the definition yet again as being the requirement that there exist an affine function g: $\mathbb{R} \to \mathbb{R}$ with $g(\xi) = f(\xi)$ such that

$$f(x) - g(x) = o(x - \xi) \qquad (x \to \xi)$$

The geometric interpretation of this condition is that the graphs $y = f(x)$ and $y = g(x)$ are tangent where $x = \xi$.



The graph of an affine function is a straight line and so our analytic definition of differentiability corresponds to the geometric idea that it be possible to draw a tangent line to the graph of f where $x = \xi$.

We are now in a position to define differentiability in the general case when m and n are not necessarily equal to one. We say that $f\colon A \to \mathbb{R}^m$ is *differentiable* at $\boldsymbol{\xi}$ if and only if there exists an affine function $g\colon A \to \mathbb{R}^m$ with $g(\boldsymbol{\xi}) = f(\boldsymbol{\xi})$ such that

$$f(\mathbf{x}) - g(\mathbf{x}) = o(\mathbf{x} - \boldsymbol{\xi}) \qquad (\mathbf{x} \to \boldsymbol{\xi}).$$

One cannot, of course, divide by a vector and $o(\mathbf{x} - \boldsymbol{\xi})$ in this equation denotes a quantity which tends to zero when divided by the real number $\|\mathbf{x} - \boldsymbol{\xi}\|$. The criterion can therefore be rewritten as

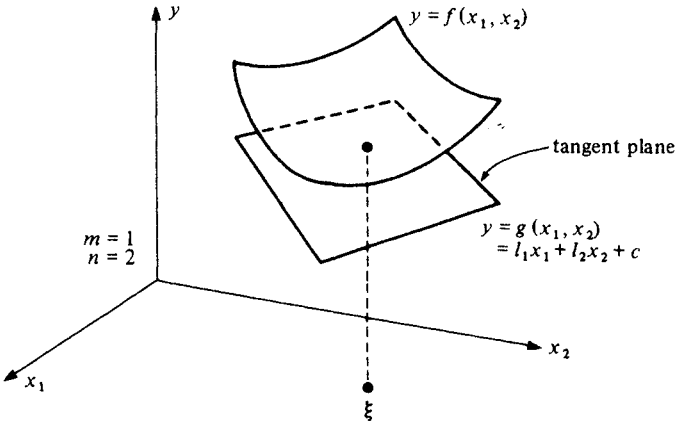
$$\lim_{\mathbf{x} \to \boldsymbol{\xi}} \frac{\|f(\mathbf{x}) - g(\mathbf{x})\|}{\|\mathbf{x} - \boldsymbol{\xi}\|} = 0.$$

As in the $m = n = 1$ case, the geometric interpretation is that the graphs $\mathbf{y} = f(\mathbf{x})$ and $\mathbf{y} = g(\mathbf{x})$ are tangent at $\mathbf{x} = \boldsymbol{\xi}$. The graph of an affine function is a *flat*. (See §18.18 and exercise 18.25(4).) The definition therefore corresponds to the geometric idea that it be possible to draw a *tangent flat* to the graph of f where $\mathbf{x} = \boldsymbol{\xi}$.

In the case $m = n = 1$, a tangent flat at $x = \xi$ is simply a straight line as illustrated above. In example 18.24(ii) we saw that an affine function $g\colon \mathbb{R}^2 \to \mathbb{R}^1$ is defined by an equation of the form

$$y = l_1 x_1 + l_2 x_2 + c$$

and its graph is therefore a plane in $\mathbb{R}^3$. In the case $m = 1$ and $n = 2$ it therefore follows that a tangent flat at $(x_1, x_2) = (\xi_1, \xi_2)$ is a plane as in the diagram below.

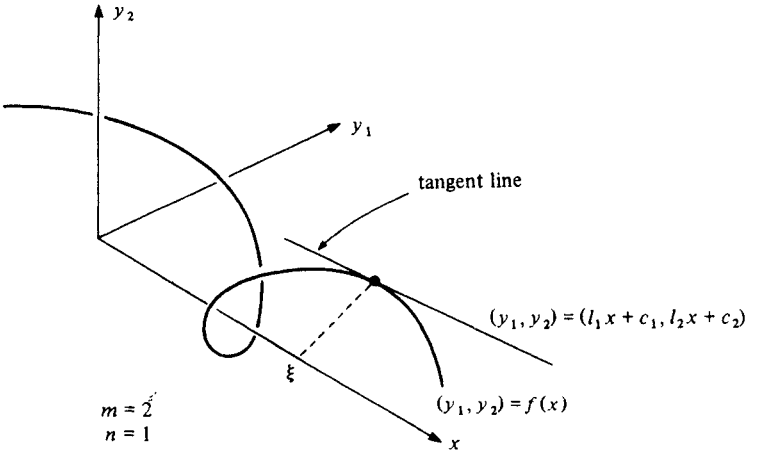


In example 18.24(iii) we saw that an affine function g: $\mathbb{R}^1 \to \mathbb{R}^2$ is defined by two equations of the form

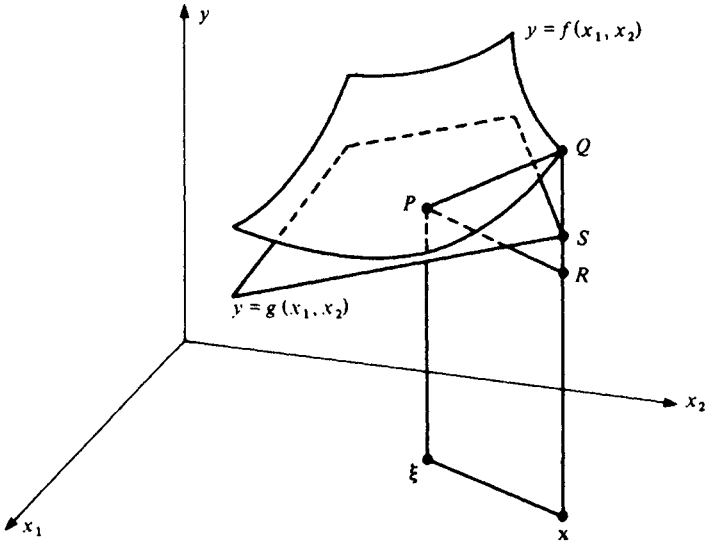
$$y_1 = l_1x_1 + c_1$$

$$y_2 = l_2x_2 + c_2$$

and its graph is therefore a line in $\mathbb{R}^3$. In the case $m = 2$ and $n = 1$, it therefore follows that a tangent flat at $x = \xi$ is a line as indicated below.



In §10.1 we gave a geometric justification for our identification of the idea of differentiability with the existence of a tangent line in the case $m = n = 1$. We now give a similar justification for the case $m = 1$ and $n = 2$.



In the diagram, Q lies on the surface $y = f(x_1, x_2)$ and S on the surface $y = g(x_1, x_2) = l_1 x_1 + l_2 x_2 + c$. The point R is at the same height as P. The condition for $y = g(x_1, x_2)$ to be tangent to $y = f(x_1, x_2)$ at P is that

$$\lim_{\mathbf{x} \to \xi} \left\{ \frac{QR}{PR} - \frac{SR}{PR} \right\} = 0.$$

But $PR = \|\mathbf{x} - \boldsymbol{\xi}\|$ and $|QR - SR| = |QS| = |f(\mathbf{x}) - g(\mathbf{x})|$. Our geometrically derived criterion therefore reduces to the condition

$$\lim_{\mathbf{x} \to \xi} \frac{|f(\mathbf{x}) - g(\mathbf{x})|}{\|\mathbf{x} - \boldsymbol{\xi}\|} = 0$$

which is the definition of differentiability in the case $m = 1$ and $n = 2$.

19.13 Theorem Let $f: A \to \mathbb{R}^m$ where A contains all $\mathbf{x} \in \mathbb{R}^n$ which satisfy $\|\mathbf{x} - \boldsymbol{\xi}\| < \Delta$. If f is differentiable at $\boldsymbol{\xi}$, then f is continuous at $\boldsymbol{\xi}$.

Proof From the definition of differentiability there exists an affine function $g: \mathbb{R}^n \to \mathbb{R}^m$ with $g(\boldsymbol{\xi}) = f(\boldsymbol{\xi})$ such that

$$\lim_{\mathbf{x} \to \xi} \frac{\|f(\mathbf{x}) - g(\mathbf{x})\|}{\|\mathbf{x} - \boldsymbol{\xi}\|} = 0$$

It follows (as in theorem 10.6) that

$$\|f(\mathbf{x}) - g(\mathbf{x})\| \to 0 \text{ as } \mathbf{x} \to \boldsymbol{\xi}.$$

But g is necessarily continuous at $\boldsymbol{\xi}$ (example 18.34). Hence

$$\begin{aligned}\|f(\mathbf{x})-f(\boldsymbol{\xi})\| &= \|f(\mathbf{x})-g(\boldsymbol{\xi})\| \\ &= \|f(\mathbf{x})-g(\mathbf{x})+g(\mathbf{x})-g(\boldsymbol{\xi})\| \\ &\leqslant \|f(\mathbf{x})-g(\mathbf{x})\|+\|g(\mathbf{x})-g(\boldsymbol{\xi})\| \\ &\to 0+0 \text{ as } \mathbf{x}\to\boldsymbol{\xi}.\end{aligned}$$

Thus $f(\mathbf{x})\to f(\boldsymbol{\xi})$ as $\mathbf{x}\to\boldsymbol{\xi}$ and so f is continuous at $\boldsymbol{\xi}$.

19.14 Derivatives

In this section we shall need the ideas introduced in §18.22. In particular, we shall find it convenient to replace $\mathbf{x}$ and $\mathbf{y}$ by *column* vectors. To emphasise this substitution we shall write x and y where previously we have written $\mathbf{x}$ and $\mathbf{y}$.

An affine function $g\colon \mathbb{R}^n\to\mathbb{R}^m$ is defined by an equation of the form

$$y = g(x) = Lx + c$$

where L is an $m\times n$ matrix and c is an $m\times 1$ column vector. If $g(\xi)=f(\xi)$, we may rewrite this equation in the form

$$y = g(x) = L(x-\xi)+f(\xi).$$

It follows that f is differentiable at ξ if and only if there exists an $m\times n$ matrix L such that

$$f(x)-f(\xi)-L(x-\xi) = o(x-\xi) \qquad (x\to\xi)$$

or, what is the same thing,

$$f(\xi+h)-f(\xi)-Lh = o(h) \qquad (h\to 0).$$

These equations are identical in form to the second and third equations of §19.12 except that L replaces l. This correspondence is very significant but it must not be forgotten that, in the current equations, the small letters all represent column *vectors* and L represents a *matrix* whereas in §19.12 all the letters in the equations represent real numbers. Note in particular that the real number l in §19.12 is the derivative $f'(\xi)$ of f at ξ. This leads us to a natural definition for the derivative of a differentiable function in the general case. If f is differentiable at ξ, we define its *derivative* at ξ by

$$f'(\xi) = L.$$

Sometimes the notation $Df(\xi)$ is used instead.

The derivative of f at ξ is therefore an $m\times n$ *matrix* in the general case. A number of advantages follow from the use of this definition. The tangent flat at ξ, for example, is given by the equation

$$y = f(\xi)+f'(\xi)(x-\xi)$$

exactly as in the case $m=n=1$.

How is the derivative of a vector function calculated? We begin to answer this question with the following theorem which connects the idea of a derivative introduced above with that of a directional derivative introduced earlier.

19.15 Theorem Suppose that $f: A \to \mathbb{R}^m$ where A contains all $x \in \mathbb{R}^n$ satisfying $\|x - \xi\| < \Delta$. If f is differentiable at ξ, then all directional derivatives of f at ξ exist and

$$D_u f(\xi) = f'(\xi)u.$$

Proof We have that

$$\frac{\|f(x) - f(\xi) - f'(\xi)(x - \xi)\|}{\|x - \xi\|} \to 0$$

as x approaches ξ and, in particular, as x approaches ξ along the straight line through ξ in the direction u. Writing $x = \xi + tu$, we obtain that

$$\lim_{t \to 0} \left\| \frac{f(\xi + tu) - f(\xi)}{t} - f'(\xi)u \right\| = 0$$

and hence $D_u f(\xi) = f'(\xi)u$.

With the assumptions of the preceding theorem, $f(x)$ denotes an $m \times 1$ column vector. The formula

$$f(x) = \begin{pmatrix} f^1(x) \\ f^2(x) \\ \vdots \\ f^m(x) \end{pmatrix}$$

therefore determines m real-valued functions $f^1, f^2, \ldots, f^m$ which we call the *component* functions of f. In the next theorem we explain how the entries in the matrix

$$f'(\xi) = \begin{pmatrix} l_{11} & l_{12} \ldots l_{1n} \\ l_{21} & l_{22} \ldots l_{2n} \\ \vdots & \\ l_{m1} & l_{m2} \ldots l_{mn} \end{pmatrix}$$

can be calculated from the partial derivatives of these component functions. (An incidental corollary is that the derivative of a differentiable function is uniquely defined.)

19.16 Theorem With the assumptions of the preceding theorem,

$$f'(\xi) = \begin{pmatrix} f_1^1(\xi) & f_2^1(\xi) \dots f_n^1(\xi) \\ f_1^2(\xi) & f_2^2(\xi) \dots f_n^2(\xi) \\ \vdots & \\ f_1^m(\xi) & f_2^m(\xi) \dots f_n^m(\xi) \end{pmatrix}$$

Proof We shall only prove that $l_{12} = f_2^1(\xi)$. We take $u = (0, 1, 0, \dots, 0)$. Then, by exercise 19.11(5),

$$D_u f(\xi) = \begin{pmatrix} D_u f^1(\xi) \\ D_u f^2(\xi) \\ \vdots \\ D_u f^m(\xi) \end{pmatrix} = \begin{pmatrix} f_2^1(\xi) \\ f_2^2(\xi) \\ \vdots \\ f_2^m(\xi) \end{pmatrix}$$

Also

$$f'(\xi)u = \begin{pmatrix} l_{11} & l_{12} \dots l_{1n} \\ l_{21} & l_{22} \dots l_{2n} \\ \vdots & \\ l_{m1} & l_{m2} \dots l_{mn} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} l_{12} \\ l_{22} \\ \vdots \\ l_{m2} \end{pmatrix}$$

Comparing the first co-ordinates of each column and appealing to theorem 19.15, we obtain that $l_{12} = f_2^1(\xi)$.

If $y = f(x)$, it is natural to use the notation

$$\frac{\mathrm{d}y}{\mathrm{d}x}$$

for the derivative of a vector function (with our usual reservations about the ambiguous use of x). The formula of theorem 19.16 then takes the pleasing and easily remembered form

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \begin{pmatrix} \dfrac{\partial y_1}{\partial x_1} & \dfrac{\partial y_1}{\partial x_2} \dots \dfrac{\partial y_1}{\partial x_n} \\ \dfrac{\partial y_2}{\partial x_1} & \dfrac{\partial y_2}{\partial x_2} \dots \dfrac{\partial y_2}{\partial x_n} \\ \vdots & \\ \dfrac{\partial y_m}{\partial x_1} & \dfrac{\partial y_m}{\partial x_2} \dots \dfrac{\partial y_m}{\partial x_n} \end{pmatrix}$$

We have seen that the derivative of a differentiable vector function f is a matrix whose entries consist of all partial derivatives of the co-ordinates of f. It is important, however, to bear in mind that the mere existence of this matrix does *not* guarantee that f is differentiable. The function of exercise 19.11(6) provides a suitable example. All its partial derivatives exist at **O** but the function

is not even continuous at **O** and hence is certainly not differentiable at **O** (theorem 19.13). Thus, although the matrix of partial derivatives can be written down, it does not help very much since the function simply does not have a tangent flat at **O**.

How can one tell whether a vector function is differentiable? The following theorem provides a criterion which is adequate for most practical purposes.

19.17 Theorem Suppose that all partial derivatives of f are *continuous* at ξ. Then f is differentiable at ξ.

Proof To avoid cumbersome notation, we consider only the case $m = 1$ and $n = 2$. From the mean value theorem (11.6) we have that

$$\begin{aligned} f(x_1, x_2) - f(\xi_1, \xi_2) &= \{f(x_1, x_2) - f(\xi_1, x_2)\} + \{f(\xi_1, x_2) - f(\xi_1, \xi_2)\} \\ &= f_1(u_1, x_2)(x_1 - \xi_1) + f_2(\xi_1, u_2)(x_2 - \xi_2) \end{aligned}$$

where u_1 lies between x_1 and ξ_1 while u_2 lies between x_2 and ξ_2. Thus $|u_1 - \xi_1| < |x_1 - \xi_1| \leqslant \|x - \xi\|$ and so $u_1 \to \xi_1$ as $x \to \xi$. It follows that $(u_1, x_2) \to (\xi_1, \xi_2)$ as $x \to \xi$. Similarly $(\xi_1, u_2) \to (\xi_1, \xi_2)$ as $x \to \xi$. Since f_1 and f_2 are continuous at ξ, we deduce that

$$f_1(u_1, x_2) \to f_1(\xi_1, \xi_2) \text{ as } x \to \xi,$$

$$f_2(\xi_1, u_2) \to f_2(\xi_1, \xi_2) \text{ as } x \to \xi$$

(a conclusion which requires the vector analogue of theorem 8.17).

We now show that

$$f(x) - f(\xi) - L(x - \xi) = o(x - \xi) \quad (x \to \xi)$$

with $L = (f_1(\xi), f_2(\xi))$. This follows from the fact that

$$\begin{aligned} &|f(x) - f(\xi) - f_1(\xi)(x_1 - \xi_1) - f_2(\xi)(x_2 - \xi_2)| \\ &\quad = |(f_1(u_1, x_2) - f_1(\xi_1, \xi_2))(x_1 - \xi_1) + (f_2(\xi_1, u_2) \\ &\quad - f_2(\xi_1, \xi_2))(x_2 - \xi_2)| \leqslant E\|x - \xi\| \end{aligned}$$

by the Cauchy–Schwarz inequality, where

$$\begin{aligned} E &= \{(f_1(u_1, x_2) - f_1(\xi_1, \xi_2))^2 + (f_2(\xi_1, u_2) - f_2(\xi_1, \xi_2))^2\}^{1/2} \\ &\to \{0^2 + 0^2\}^{1/2} = 0 \text{ as } x \to \xi. \end{aligned}$$

19.18 Examples

(i) Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(x_1, x_2) = x_1^2 x_2 + x_2^3 x_1$. Then

$$\left.\begin{aligned} f_1(x_1, x_2) &= 2x_1 x_2 + x_2^3 \\ f_2(x_1, x_2) &= x_1^2 + 3x_2^2 x_1. \end{aligned}\right\}$$

These partial derivatives are polynomials and hence are continuous everywhere (proposition 18.38). Thus f is differentiable at every point. The derivative at the point (2, 1) is the 1×2 matrix

$$\begin{aligned} f'(2, 1) &= (f_1(2, 1), f_2(2, 1)) \\ &= (5, 10). \end{aligned}$$

Various facts about the behaviour of f at (2, 1) may be deduced from this information. For example, the slope (or rate of increase) of f at the point (2, 1) in the direction (4, 3) is given by

$$D_u f(1, 2)$$

where u is the unit vector which points in the same direction as (4, 3). From theorem 19.15 we have that

$$\begin{aligned} D_u f(1, 2) &= f'(1, 2)u \\ &= (5, 10)\begin{pmatrix} 4/5 \\ 3/5 \end{pmatrix} = 4 + 6 = 10. \end{aligned}$$

We can also obtain the equation of the tangent plane to the surface $y = x_1^2 x_2 + x_2^3 x_1$ where $(x_1, x_2) = (1, 2)$. This is given by writing $(\xi_1, \xi_2) = (1, 2)$ in the formula

$$y = f(\xi) + f'(\xi)(x - \xi).$$

This yields

$$y = 6 + (5, 10)\begin{pmatrix} x_1 - 1 \\ x_2 - 2 \end{pmatrix}$$

i.e.

$$y - 6 = 5(x_1 - 1) + 10(x_2 - 2).$$

(ii) Let $f\colon \mathbb{R}^3 \to \mathbb{R}$ be defined by $f(x_1, x_2, x_3) = x_1 x_2^2 + x_2 x_3^3 + x_3 x_1^4$. Then

$$\left.\begin{aligned} f_1(x_1, x_2, x_3) &= x_2^2 + 4x_3 x_1^3 \\ f_2(x_1, x_2, x_3) &= 2x_1 x_2 + x_3^3 \\ f_3(x_1, x_2, x_3) &= 3x_2 x_3^2 + x_1^4. \end{aligned}\right\}$$

These partial derivatives are continuous at every point and hence f is differentiable everywhere. The derivative at (1, 1, 1) is the 1×3 matrix

$$f'(1, 1, 1) = (5, 3, 4).$$

The slope (or rate of increase) of f at (1, 1, 1) in the direction (2, 1, 2) is equal to

$$f'(1, 1, 1)u = (5, 3, 4)\begin{pmatrix} 2/3 \\ 1/3 \\ 2/3 \end{pmatrix} = 7$$

(where u is the unit vector which points in the same direction as $(2, 1, 2)$).

The tangent flat to $y = x_1x_2^2 + x_2x_3^3 + x_3x_1^4$ where $(x_1, x_2, x_3) = (1, 1, 1)$ is the hyperplane

$$y = 3 + (5, 3, 4)\begin{pmatrix} x_1 - 1 \\ x_2 - 1 \\ x_3 - 1 \end{pmatrix}$$

i.e.

$$y - 3 = 5(x_1 - 1) + 3(x_2 - 1) + 4(x_3 - 1).$$

(iii) Let $f\colon \mathbb{R} \to \mathbb{R}^3$ be defined by

$$f(x) = \begin{pmatrix} x \\ x^2 \\ x^3 \end{pmatrix}.$$

The component functions $f^1\colon \mathbb{R} \to \mathbb{R}$, $f^2\colon \mathbb{R} \to \mathbb{R}$ and $f^3\colon \mathbb{R} \to \mathbb{R}$ are given by

$$\left.\begin{aligned} f^1(x) &= x \\ f^2(x) &= x^2 \\ f^3(x) &= x^3 \end{aligned}\right\}$$

and hence

$$\left.\begin{aligned} f_1^1(x) &= 1 \\ f_1^2(x) &= 2x \\ f_1^3(x) &= 3x^2. \end{aligned}\right\}$$

These functions are all continuous everywhere and hence f is differentiable at every point. Its derivative at -1 is the 3×1 matrix

$$f'(-1) = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}.$$

The graph of f is determined by the system of three equations

$$\left.\begin{aligned} y_1 &= x \\ y_2 &= x^2 \\ y_3 &= x^3. \end{aligned}\right\}$$

The tangent flat where $x = -1$ is given by

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix} + \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}(x+1)$$

and hence is the straight line determined by the equations

$$\left.\begin{aligned} y_1 &= x \\ y_2 &= -2x - 1 \\ y_3 &= 3x + 2. \end{aligned}\right\}$$

(iv) Let $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ be defined by

$$f(x_1, x_2) = \begin{pmatrix} x_1^2 + x_2^2 \\ x_1^2 - x_2^2 \end{pmatrix}.$$

The component functions $f^1\colon \mathbb{R}^2 \to \mathbb{R}$ and $f^2\colon \mathbb{R}^2 \to \mathbb{R}$ are then given by

$$\begin{aligned} f^1(x_1, x_2) &= x_1^2 + x_2^2 \\ f^2(x_1, x_2) &= x_1^2 - x_2^2 \end{aligned}$$

and hence

$$\begin{aligned} f_1^1(x_1, x_2) &= 2x_1 \qquad f_2^1(x_1, x_2) = 2x_2 \\ f_1^2(x_1, x_2) &= 2x_1 \qquad f_2^2(x_1, x_2) = -2x_2. \end{aligned}$$

These functions are continuous everywhere and hence f is differentiable at every point. Its derivative at $(1, 1)$ is the 2×2 matrix

$$f'(1, 1) = \begin{pmatrix} 2 & 2 \\ 2 & -2 \end{pmatrix}.$$

The graph of f is determined by the two equations

$$\left.\begin{aligned} y_1 &= x_1^2 + x_2^2 \\ y_2 &= x_1^2 - x_2^2. \end{aligned}\right\}$$

The tangent flat where $(x_1, x_2) = (1, 1)$ is given by

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix} + \begin{pmatrix} 2 & 2 \\ 2 & -2 \end{pmatrix}\begin{pmatrix} x_1 - 1 \\ x_2 - 1 \end{pmatrix}$$

i.e.

$$\left.\begin{aligned} y_1 - 2 &= 2(x_1 - 1) + 2(x_2 - 1) \\ y_2 &= 2(x_1 - 1) - 2(x_2 - 1). \end{aligned}\right\}$$

19.19 Gradient

As though there were not enough notation associated with vector functions, there is a further notation which is used for the derivative of a real-valued function. A derivative $f'(\xi)$ in this case is a $1 \times n$ matrix, i.e. a row vector. When matrix notation is not being systematically used especially in applied mathematics, one sometimes writes

$$\nabla f(\boldsymbol{\xi}) = f'(\xi)$$

and refers to $\nabla f(\boldsymbol{\xi})$ as the *gradient* of f at $\boldsymbol{\xi}$.

The geometric interpretation of the gradient (i.e. the derivative) of a real-valued function of a vector variable is worth noting. Recall that

$$y - f(\xi) = f'(\xi)(x - \xi)$$

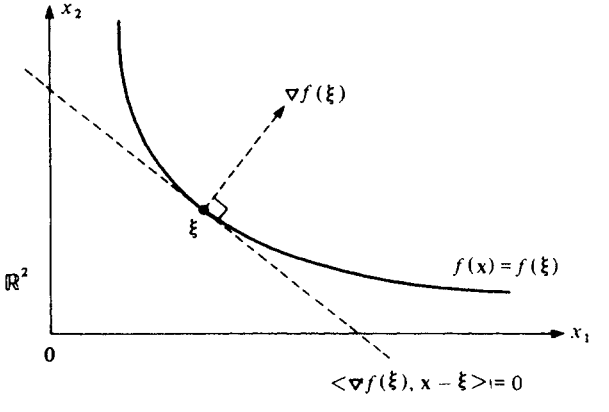
is the equation of the hyperplane in $\mathbb{R}^{n+1}$ which is the tangent to $y = f(x)$ where $x = \xi$. It follows that

$$f'(\xi)(x - \xi) = 0$$

is the equation of the hyperplane in $\mathbb{R}^n$ which is the tangent to the contour $f(x) = f(\xi)$ where $x = \xi$. Rewriting the equation $f'(\xi)(x - \xi) = 0$ in the form

$$\langle \nabla f(\boldsymbol{\xi}), \mathbf{x} - \boldsymbol{\xi} \rangle = 0$$

and referring to §18.15, we observe that $\nabla f(\boldsymbol{\xi})$ is a *normal* to the contour $f(\mathbf{x}) = f(\boldsymbol{\xi})$ at $\mathbf{x} = \boldsymbol{\xi}$.



This provides a geometric interpretation for the direction in which $\nabla f(\boldsymbol{\xi})$ points. Its length can also be interpreted geometrically. The slope (or rate of increase) of f at $\boldsymbol{\xi}$ in the direction of the unit vector $\mathbf{u}$ is given by

$$D_{\mathbf{u}} f(\boldsymbol{\xi}) = f'(\xi)u = \langle \nabla f(\boldsymbol{\xi}), \mathbf{u} \rangle.$$

By the Cauchy–Schwarz inequality,

$$|D_{\mathbf{u}}f(\boldsymbol{\xi})| = |\langle \nabla f(\boldsymbol{\xi}), \mathbf{u}\rangle| \leqslant \|\nabla f(\boldsymbol{\xi})\|.\|\mathbf{u}\| = \|\nabla f(\boldsymbol{\xi})\|.$$

But, taking $\mathbf{u} = \|\nabla f(\boldsymbol{\xi})\|^{-1}\nabla f(\boldsymbol{\xi})$, we obtain that

$$D_{\mathbf{u}}f(\boldsymbol{\xi}) = \|\nabla f(\boldsymbol{\xi})\|.$$

Thus $\nabla f(\boldsymbol{\xi})$ points in the direction in which the slope of f at $\boldsymbol{\xi}$ is *greatest* and the magnitude of the slope in this direction is equal to $\|\nabla f(\boldsymbol{\xi})\|$.

19.20 Example Consider the function $f\colon \mathbb{R}^2 \to \mathbb{R}$ of example 19.18(i). We have that

$$\nabla f(2, 1) = f'(2, 1) = (5, 10).$$

It follows that (5, 10) is a normal to the contour $x_1^2x_2 + x_2^3x_1 = 6$ where $(x_1, x_2) = (2, 1)$. Also (5, 10) is the direction in which f increases fastest at (2,1) and the rate of increase in this direction is $\{5^2 + 10^2\}^{1/2} = 5\sqrt{5}$.

19.21 Manipulation of derivatives

For real-valued functions of a real variable, the following formulae are familiar.

$$\text{(I)} \quad \frac{d}{dx}(Lx + c) = L$$

$$\text{(II)} \quad \frac{d}{dx}(Lx^2) = 2Lx$$

$$\text{(III)} \quad \frac{d}{dx}(Ly + Mz) = L\frac{dy}{dx} + M\frac{dz}{dx}$$

$$\text{(IV)} \quad \frac{d}{dx}(yz) = z\frac{dy}{dx} + y\frac{dz}{dx}$$

$$\text{(V)} \quad \frac{dz}{dx} = \frac{dz}{dy}\cdot\frac{dy}{dx}$$

$$\text{(VI)} \quad \frac{dy}{dx} = \left(\frac{dx}{dy}\right)^{-1}.$$

The range of validity of these formulae was discussed in chapter 10. It is a very convenient fact that versions of all these formulae remain valid when the small letters are re-interpreted as column vectors and the large letters as matrices, *provided* that the formula obtained after this re-interpretation makes sense. With this re-interpretation, for example, formula (V) asserts that one matrix is the

product of two other matrices and formula (VI) asserts that one matrix is the inverse of another. (The latter assertion, of course, makes no sense unless

$$\frac{dy}{dx}$$

is a square, non-singular matrix which requires in particular that x and y have the same number of co-ordinates.)

We leave further consideration of (V) and (VI) until the next section and focus attention for the moment on the other formulae. As far as these are concerned, (I) and (III) are straightforward to interpret and easy to prove (exercise 19.26(4)). Formulae (II) and (IV), however, make no immediate sense when x, y and z are interpreted as column vectors. We therefore re-arrange the formulae slightly so that they do make sense and obtain

$$\text{(II)}' \quad \frac{d}{dx}(x^{\mathrm{T}}Ax) = x^{\mathrm{T}}(A^{\mathrm{T}} + A)$$

$$\text{(IV)}' \quad \frac{d}{dx}(y^{\mathrm{T}}z) = z^{\mathrm{T}}\frac{dy}{dx} + y^{\mathrm{T}}\frac{dz}{dx}.$$

As an example, we shall deduce (II)′ from (IV)′. The proof of the latter formula is left as an exercise (19.26(4)).

19.22 Example Observe to begin with that (II)′ makes no sense unless A is a *square* matrix. From formula (IV)′ and (I),

$$\frac{d}{dx}x^{\mathrm{T}}Ax = \frac{d}{dx}x^{\mathrm{T}}(Ax) = (Ax)^{\mathrm{T}}\frac{dx}{dx} + x^{\mathrm{T}}\frac{d}{dx}(Ax)$$

$$= x^{\mathrm{T}}A^{\mathrm{T}}I + x^{\mathrm{T}}A = x^{\mathrm{T}}(A^{\mathrm{T}} + A).$$

Here I denotes the *identity matrix*, i.e.

$$I = \begin{pmatrix} 1 & 0 \ldots 0 \\ 0 & 1 \ldots 0 \\ \vdots & \\ 0 & 0 \ldots 1 \end{pmatrix}.$$

Note that in the special case when A is *symmetric*, i.e. $A^{\mathrm{T}} = A$, we have that

$$\frac{d}{dx}(x^{\mathrm{T}}Ax) = 2Ax.$$

19.23 Chain rule

As we know from §10.12 that formula (V), i.e.

$$\frac{dz}{dx} = \frac{dz}{dy}\cdot\frac{dy}{dx}$$

is the formula for differentiating a *composite function.* To justify this formula in the vector case, we therefore need to establish the following result. (See theorem 10.13.)

19.24 Proposition Suppose that g is differentiable at ξ and f is differentiable at $\eta = g(\xi)$. Then $f \circ g$ is differentiable at ξ and

$$(f \circ g)'(\xi) = f'(\eta)g'(\xi).$$

Proof We are given that there exist matrices $f'(\eta)$ and $g'(\xi)$ for which

$$f(y) - f(\eta) - f'(\eta)(y - \eta) = o(y - \eta) \quad (y \to \eta)$$

$$g(x) - g(\xi) - g'(\xi)(x - \xi) = o(x - \xi) \quad (x \to \xi).$$

Writing $y = g(x)$ and $\eta = g(\xi)$ in the first equation, we obtain that

$$f(g(x)) - f(g(\xi)) - f'(\eta)(g(x) - g(\xi)) = o(g(x) - g(\xi))$$

and hence

$$f(g(x)) - f(g(\xi)) - f'(\eta)\{g'(\xi)(x - \xi) + o(x - \xi)\}$$
$$= o\{g'(\xi)(x - \xi) + o(x - \xi)\}.$$

It follows that

$$f(g(x)) - f(g(\xi)) - f'(\eta)g'(\xi)(x - \xi) = o(x - \xi) \quad (x \to \xi)$$

and so $f \circ g$ is differentiable at ξ with derivative $f'(\eta)g'(\xi)$. As in theorem 10.13, the final step is not entirely straightforward but we prefer not to hold up the discussion by presenting a detailed proof at this stage.

Written in full, formula (V) takes the form

$$\begin{pmatrix} \frac{\partial z_1}{\partial x_1} & \frac{\partial z_1}{\partial x_2} & \cdots & \frac{\partial z_1}{\partial x_n} \\ \frac{\partial z_2}{\partial x_1} & \frac{\partial z_2}{\partial x_2} & \cdots & \frac{\partial z_2}{\partial x_n} \\ \vdots & \vdots & & \\ \frac{\partial z_m}{\partial x_1} & \frac{\partial z_m}{\partial x_2} & \cdots & \frac{\partial z_m}{\partial x_n} \end{pmatrix} = \begin{pmatrix} \frac{\partial z_1}{\partial y_1} & \cdots & \frac{\partial z_1}{\partial y_p} \\ \frac{\partial z_2}{\partial y_1} & \cdots & \frac{\partial z_2}{\partial y_p} \\ \vdots & & \\ \frac{\partial z_m}{\partial y_1} & \cdots & \frac{\partial z_m}{\partial y_p} \end{pmatrix} \begin{pmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \cdots & \frac{\partial y_1}{\partial x_n} \\ \vdots & & & \\ \frac{\partial y_p}{\partial x_1} & \frac{\partial y_p}{\partial x_2} & \cdots & \frac{\partial y_p}{\partial x_n} \end{pmatrix}$$

from which we deduce that

$$\frac{\partial z_i}{\partial x_j} = \frac{\partial z_i}{\partial y_1}\frac{\partial y_1}{\partial x_j} + \frac{\partial z_i}{\partial y_2}\frac{\partial y_2}{\partial x_j} + \ldots + \frac{\partial z_i}{\partial y_p}\frac{\partial y_p}{\partial x_j}.$$

This formula is called the *chain rule* and is often very useful in calculating partial derivatives.

19.25 Examples

(i) Let f: $\mathbb{R}^2 \to \mathbb{R}^1$ and g: $\mathbb{R}^1 \to \mathbb{R}^2$ be defined by

$$z = f(y_1, y_2) = y_1^2 y_2 + y_2^3 y_1 ; y = g(x) = \begin{pmatrix} x^2 \\ x^3 \end{pmatrix}.$$

From formula (V)

$$\frac{dz}{dx} = \frac{dz}{dy}\cdot\frac{dy}{dx} = (2y_1y_2 + y_2^3, y_1^2 + 3y_2^2y_1)\begin{pmatrix} 2x \\ 3x^2 \end{pmatrix}.$$

This has to be calculated when $y = g(x)$, i.e. when $y_1 = x^2$ and $y_2 = x^3$. Thus

$$\frac{dz}{dx} = (2x^5 + x^9, x^4 + 3x^8)\begin{pmatrix} x^2 \\ x^3 \end{pmatrix} = 7x^6 + 11x^{10}.$$

The same result can be easily obtained directly in this case because $f \circ g$: $\mathbb{R}^1 \to \mathbb{R}^1$ is given by the simple formula $f \circ g(x) = (x^2)^2x^3 + (x^3)^3x = x^7 + x^{11}$.

(ii) Suppose that $w = xy^2z$ where

$$x = r + 2s + 3t$$

$$y = 2r + 3s + t$$

$$z = 3r + s + t.$$

We can calculate the partial derivative of w with respect to r, s or t using the chain rule. For example

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial s} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial s}.$$

It is as well to appreciate that every symbol in this formula is used ambiguously. This can naturally lead to confusion unless one is quite clear about what everything means. It is sometimes helpful to rewrite the equation in the form

$$\left(\frac{\partial w}{\partial s}\right)_{r,t} = \left(\frac{\partial w}{\partial x}\right)_{y,z}\left(\frac{\partial x}{\partial s}\right)_{r,t} + \left(\frac{\partial w}{\partial y}\right)_{x,z}\left(\frac{\partial y}{\partial s}\right)_{r,t} + \left(\frac{\partial w}{\partial z}\right)_{x,y}\left(\frac{\partial z}{\partial s}\right)_{r,t}$$

to make it clear what is to be held constant in each differentiation. (See § 19.5.) Thus

$$\frac{\partial w}{\partial s} = y^2z^32 + 2xyz^33 + 3xy^2z^2 = 2(2r + 3s + t)^2(3r + s + t)^3$$
$$+ 6(r + 2s + 3t)(2r + 3s + t)(3r + s + t)^3$$
$$+ 3(r + 2s + 3t)(2r + 3s + t)^2(3r + s + t)^2.$$

We now turn to formula (VI), i.e.

$$\frac{dy}{dx} = \left(\frac{dx}{dy}\right)^{-1}.$$

As we know from § 12.9, this formula concerns the derivative of an *inverse function.* Assuming that f has an inverse function f^{-1}, then

$$f \circ f^{-1}(x) = x$$

(exercise 7.16(5)). If f is differentiable at ξ and f^{-1} is differentiable at $\eta = f(\xi)$, then the formula (V) yields that

$$f'(\xi)(f^{-1})'(\eta) = I$$

from which formula (VI) follows.

When considering the real case we also gave conditions under which a suitable differentiable inverse exists. It follows from theorem 12.10 that it is sufficient in the real case if f' exists and is positive on some open interval containing ξ. The same is, of course, true if the word positive is replaced by the word negative. In the particular case when f' is continuous at ξ, it is therefore enough if $f'(\xi) \neq 0$.

A similar result holds in the vector case. Note that $f'(\xi) \neq 0$ in the real case corresponds to the requirement that the affine function tangent to f at ξ has an inverse function. The condition in the vector case which corresponds to this requirement is that the $n \times n$ matrix $f'(\xi)$ be *non-singular*, i.e.

$$\det f'(\xi) \neq 0.$$

We call the determinant $\det f'(\xi)$ the *Jacobian* of f at ξ. It is an important result that a suitable differentiable inverse of f exists provided that all the first order partial derivatives of f are continuous at ξ and $\det f'(\xi) \neq 0$. However, we shall neither explain the meaning of the word 'suitable' in this context nor offer a proof of the result. A proof would require apparatus that has not been developed in this book.

19.26 Exercise

(1) Explain why all functions f: $\mathbb{R}^n \to \mathbb{R}^m$ whose co-ordinates are all polynomials are differentiable at every point. What is the appropriate result when the co-ordinates are rational functions?

(2) Explain why each of the following functions is differentiable at the given point and compute its derivative there.

(i) $f\colon \mathbb{R}^2 \to \mathbb{R}$ at $(1, 2)$ where $f(x_1, x_2) = x_1^2 + 2x_1x_2 + x_2^3$.

(ii) $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ at $(\pi, 1)$ where

$$f(x_1, x_2) = \begin{pmatrix} \cos(x_1x_2) \\ \sin(x_1x_2) \end{pmatrix}.$$

(iii) $f\colon \mathbb{R}^2 \to \mathbb{R}^3$ at $(1, 1)$ where

$$f(x_1, x_2) = \begin{pmatrix} x_1 \\ x_1 + x_2 \\ x_1x_2 \end{pmatrix}.$$

(iv) $f\colon \mathbb{R}^3 \to \mathbb{R}^2$ at $(1, 2, 3)$ where

$$f(x_1, x_2, x_3) = \begin{pmatrix} x_1x_2^2 \\ x_2^2x_3^3 \end{pmatrix}.$$

(v) $f\colon \mathbb{R} \to \mathbb{R}^3$ at 0 where

$$f(x) = \begin{pmatrix} 1 \\ x \\ x^2 \end{pmatrix}$$

For each of these functions, write down the equations for the appropriate tangent flat.

(3) What is the slope (or rate of increase) of the function f of question 2(i) at the point $(1, 2)$ in the direction $(1, -1)$? What is the direction at $(1, 2)$ in which the slope is greatest? What is the value of the slope in this direction? Write down a vector which is normal to the contour $x_1^2 + 2x_1x_2 + x_2^3 = 13$ at $(1, 2)$. Also write down the equation of the tangent line at this point.

(4) Explain the circumstances under which formulae (I), (III) and (IV)′ of §19.21 make sense when L and M are matrices and x, y, z and c are column vectors. Prove that the formulae are valid when they make sense.

(5) Suppose that $f\colon \mathbb{R} \to \mathbb{R}$ is differentiable everywhere and that the function $\phi\colon \mathbb{R}^2 \to \mathbb{R}$ is defined by $\phi(x_1, x_2) = f(x_1x_2)$. If $y = \phi(x_1, x_2)$, prove that

$$x_1 \frac{\partial y}{\partial x_1} = x_2 \frac{\partial y}{\partial x_2}$$

for all $(x_1, x_2) \in \mathbb{R}^2$ for which $f'(x_1x_2) \neq 0$.

(6) Suppose that f: $\mathbb{R}^2 \to \mathbb{R}$ is differentiable everywhere and that $y = f(x_1, x_2)$ where

$$x_1 = r\cos\theta$$

$$x_2 = r\sin\theta.$$

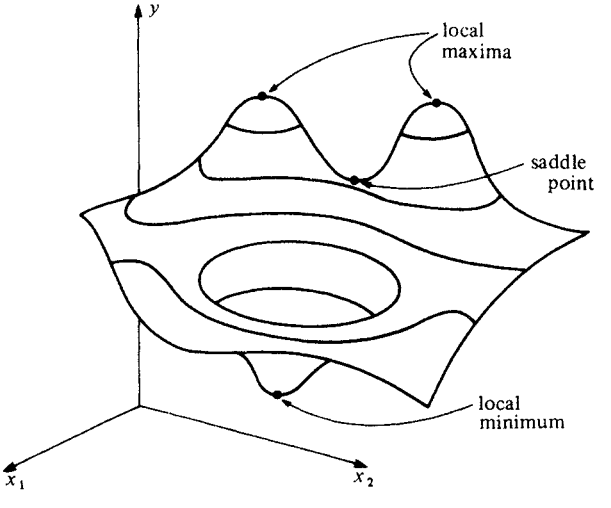
Express $\partial y/\partial x_2$ in terms of $\partial y/\partial r$ and $\partial y/\partial\theta$.

19.27 Stationary points

In §11.3 we defined a point ξ to be a *stationary point* of f if $f'(\xi) = 0$. We shall use precisely the same definition in the case of a real-valued function of a vector variable. In this case, of course, $f'(\xi)$ is a $1 \times n$ row vector and $f'(\xi) = 0$ means that all the co-ordinates of this row vector (i.e. the partial derivatives of f at ξ) are zero. In geometric terms, the fact that ξ is a stationary point means that the tangent hyperplane

$$y = f(\xi) + f'(\xi)(x - \xi)$$

reduces to $y = f(\xi)$, i.e. the tangent hyperplane is horizontal at a stationary point. The diagram shows some points on a surface $y = f(x_1, x_2)$ at which the tangent plane is horizontal.



Any local maximum or minimum at which f is differentiable is, of course, a stationary point. This follows immediately from corollary 19.9 and we quote the result as a theorem below. It is not at all uncommon, however, for stationary points in the vector case to correspond to *saddle points* as indicated in the

diagram. Less common, but not to be ignored, are other more exotic stationary points similar to those mentioned in §11.3.

19.28 Theorem Suppose that $f: A \to \mathbb{R}$ where A is a set in $\mathbb{R}^n$. If f is differentiable at $\xi \in A$ and f has a local maximum or minimum at ξ, then

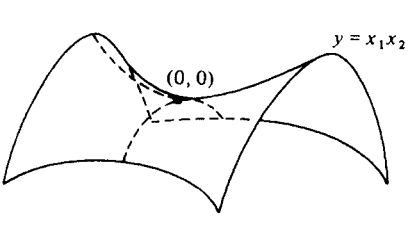
$$f'(\xi) = 0.$$

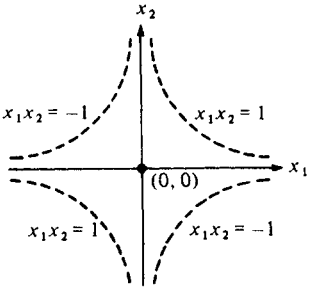
Proof This follows from corollary 19.9 and theorem 19.16.

19.29 Example Consider the function $f: \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x_1, x_2) = x_1 x_2$. Observe that

$$f'(0, 0) = (f_1(0, 0), f_2(0, 0)) = (0, 0)$$

but f does *not* have a local maximum or minimum at (0, 0). On the line $x_1 = x_2$ we have that $f(x_1, x_2) = x_1^2 > 0$ unless $x_1 = 0$ while, on the line $x_1 = -x_2$, $f(x_1, x_2) = -x_1^2 < 0$ unless $x_1 = 0$.





The two diagrams are alternative representations of the situation. The left hand diagram makes it clear why f is said to have a saddle point at (0, 0).

19.30 Example In 'least squares analysis', one is given an *inconsistent* system $Ax = b$ of linear equations and asked to find the vector or vectors x which minimises $\|Ax - b\|$. Consider

$$y = \|Ax - b\|^2 = (Ax - b)^{\mathrm{T}}(Ax - b) = u^{\mathrm{T}}u$$

where $u = Ax - b$. By formula (V) of §19.21,

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = (2u^{\mathrm{T}}I)A = 2(Ax - b)^{\mathrm{T}}A.$$

At a local minimum, the derivative must be the zero vector and so, if x minimises $\|Ax - b\|$, then

$$2(Ax-b)^{\mathrm{T}}A = 0.$$

Recalling exercise 18.25(2), we rewrite this as

$$A^{\mathrm{T}}(Ax-b) = 0.$$

i.e.

$$A^{\mathrm{T}}Ax = A^{\mathrm{T}}b.$$

If $A^{\mathrm{T}}A$ is non-singular, it follows that there is only one candidate for a minimising x, namely

$$x = (A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}b.$$

In the preceding example, the geometry of the situation makes it clear that a minimising x must exist. However, a geometrical argument is no substitute for a proof and, in any case, the geometry of the situation is seldom as transparent as in example 19.30. Some method is therefore necessary for classifying stationary points into local maxima, local minima, saddle points or other more exotic categories. Our experience with the real case would indicate an examination of the *second* derivative at a stationary point. But how is this defined?

19.31 Second derivatives

We consider only the second derivative of *real-valued* functions. The reason for this is that the derivative of a real-valued function is a vector and we know how to differentiate vector-valued functions. In order to discuss second derivatives in general, we should need first to discuss the derivative of matrix-valued functions and for this purpose the notation of matrix algebra is not adequate. (See §19.48.)

Even restricting ourselves to real-valued functions, we have a small problem since §19.14 assumes that we have a column vector to differentiate whereas $f'(x)$ is a row vector. However, this problem is easily overcome. If $\phi(x)$ is a row vector we define $\phi'(x)$ by

$$\phi'(x)^{\mathrm{T}} = \frac{d}{dx}\{\phi(x)^{\mathrm{T}}\}$$

provided the latter derivative exists. If ϕ is differentiable at ξ, we then have that

$$\phi(x)-\phi(\xi)-(x-\xi)^{\mathrm{T}}\phi'(\xi) = o(x-\xi)$$

i.e. the theory for row vectors is the same as that for column vectors except that it is necessary to write $(x-\xi)^{\mathrm{T}}\phi'(\xi)$ rather than $f'(\xi)(x-\xi)$ wherever the latter expression occurs.

With this understanding, we can now define the second derivative of a real-valued function f at the point x by

$$f''(x)^{\mathrm{T}} = \frac{d}{dx} f'(x)^{\mathrm{T}}$$

provided the latter derivative exists. Since

$$f'(x)^{\mathrm{T}} = \begin{pmatrix} f_1(x) \\ f_2(x) \\ \vdots \\ f_n(x) \end{pmatrix}$$

it follows immediately from theorem 19.16 that $f''(x)$ is the $n \times n$ matrix

$$f''(x) = \begin{pmatrix} f_{11}(x) & f_{12}(x) \ldots f_{1n}(x) \\ f_{21}(x) & f_{22}(x) \ldots f_{2n}(x) \\ \ldots\ldots\ldots\ldots\ldots\ldots \\ f_{n1}(x) & f_{n2}(x) \ldots f_{nn}(x) \end{pmatrix} \tag{1}$$

where

$$f_{ij}(x) = \frac{\partial}{\partial x_i}\{f_j(x)\} = \frac{\partial}{\partial x_i}\frac{\partial}{\partial x_j} f(x)\}.$$

It is customary to abbreviate the latter notation and to write

$$f_{ij}(x) = \frac{\partial^2}{\partial x_i \partial x_j} f(x).$$

Thus, if $y = f(x)$,

$$\frac{d^2y}{dx^2} = \begin{pmatrix} \dfrac{\partial^2 y}{\partial x_1^2} & \dfrac{\partial^2 y}{\partial x_1 \partial x_2} \cdots \dfrac{\partial^2 y}{\partial x_1 \partial x_n} \\ \dfrac{\partial^2 y}{\partial x_2 \partial x_1} & \dfrac{\partial^2 y}{\partial x_2^2} \quad \cdots \dfrac{\partial^2 y}{\partial x_2 \partial x_n} \\ \vdots \\ \dfrac{\partial^2 y}{\partial x_n \partial x_1} & \dfrac{\partial^2 y}{\partial x_n \partial x_2} \cdots \dfrac{\partial^2 y}{\partial x_n^2} \end{pmatrix}$$

and so the second derivative of a real-valued function consists of an $n \times n$ matrix whose entries consist of all second order partial derivatives of the function. From theorem 19.17 we know that a sufficient condition that f be twice differentiable at a point ξ is that all these second order partial derivatives exist and are *continuous* at ξ.

19.32 Example Consider the function $f\colon \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x_1, x_2) = x_1^2 x_2 + x_2^3 x_1$. We have that

$$\frac{\partial y}{\partial x_1} = 2x_1x_2 + x_2^3 \qquad \frac{\partial y}{\partial x_2} = x_1^2 + 3x_2^2x_1$$

and so

$$\frac{\partial^2 y}{\partial x_1^2} = 2x_2 \qquad \frac{\partial^2 y}{\partial x_1 \partial x_2} = 2x_1 + 3x_2^2$$

$$\frac{\partial^2 y}{\partial x_2 \partial x_1} = 2x_1 + 3x_2^2 \qquad \frac{\partial^2 y}{\partial x_2^2} = 6x_1x_2.$$

All these are continuous everywhere and hence f is twice differentiable at every point with

$$f''(x_1, x_2) = \begin{pmatrix} 2x_2 & 2x_1 + 3x_2^2 \\ 2x_1 + 3x_2^2 & 6x_1x_2 \end{pmatrix}.$$

It is no accident that the matrix $f''(x_1, x_2)$ in the preceding example is *symmetric*, i.e.

$$\frac{\partial^2 y}{\partial x_1 \partial x_2} = \frac{\partial^2 y}{\partial x_2 \partial x_1}.$$

19.33 Proposition If all the second order partial derivatives in the matrix (1) of §19.31 are continuous at ξ, then the $n \times n$ matrix $f''(\xi)$ is symmetric, i.e. for all i and j

$$f_{ij}(\xi) = f_{ji}(\xi).$$

19.34 Mean value theorems

The mean value theorem of §11.5 generalises immediately to the case of *real-valued* functions of a vector variable. Note, however, that it does *not* generalise to vector-valued functions. (See exercise 19.47(5).)

19.35 Theorem (mean value theorem) Let L denote the straight line segment joining $a \in \mathbb{R}^n$ and $b \in \mathbb{R}^n$ and let L^0 be the set obtained by deleting a and b from L. If f is a real-valued function which is continuous at each point of L and differentiable at each point of L^0, then

$$f(b) - f(a) = f'(\xi)(b - a)$$

for some $\xi \in L^0$.

Proof Define F: $[0, 1] \to \mathbb{R}$ by

$$F(t) = f((1 - t)a + tb) \; \mathbb{R} \; (0 \leqslant t \leqslant 1).$$

Then F is continuous on $[0, 1]$ and differentiable on $(0, 1)$. (See exercise 18.20(5).) By theorem 11.6

$$\frac{F(1)-F(0)}{1-0} = F'(\tau)$$

for some $\tau \in (0, 1)$. Thus $f(b)-f(a) = F(1)-F(0) = F'(\tau)$. It remains to express $F'(\tau)$ in terms of the function f. Write $y = f(x)$ and $x = (1-t)a + tb$. Then

$$F'(t) = \frac{dy}{dt} = \frac{dy}{dx}\cdot\frac{dx}{dt} = f'(x)(b-a).$$

Writing $\xi = (1-\tau)a + \tau b$, it follows that

$$f(b)-f(a) = F'(\tau) = f'(\xi)(b-a).$$

In chapter 11, we went on to prove Taylor's theorem which, as we pointed out, may be regarded as the nth order mean value theorem. This also generalises to the case of *real-valued* functions but we consider only the case $n = 2$ since we have not discussed derivatives of higher order.

19.36 Theorem (Taylor's theorem of order 2) Suppose that f is a real-valued function which is twice differentiable at each point $x \in \mathbb{R}^n$ which satisfies $\|x-\xi\| < \Delta$. Then for any such x,

$$f(x) = f(\xi) + \frac{1}{1!}f'(\xi)(x-\xi) + \frac{1}{2!}(x-\xi)^{\mathrm{T}}f''(\eta)(x-\xi)$$

for some η on the line segment L^0 between x and ξ.

Proof We apply theorem 11.10 to the function F: $[0, 1] \to \mathbb{R}$ defined by $F(t) = f((1-t)\xi + tx)$ $(0 \leqslant t \leqslant 1)$ and obtain

$$F(1) = F(0) + \frac{1}{1!}F'(0) + \frac{1}{2!}F''(\tau)$$

for some τ satisfying $0 < \tau < 1$. We have that $F(1) = f(x)$, $F(0) = f(\xi)$ and, as in the previous theorem,

$$F'(t) = f'((1-t)\xi + tx)(x-\xi)$$

so that $F'(0) = f'(\xi)(x-\xi)$. It remains to express $F''(\tau)$ in terms of f. This requires a little care with transposition signs. (See exercise 18.25(2).) Observe that

$$F''(t)^{\mathrm{T}} = \frac{d}{dt}\{F'(t)\}^{\mathrm{T}}$$

$$= \frac{d}{dt}\{f'((1-t)\xi + tx)(x-\xi)\}^{\mathrm{T}}$$

$$= \frac{d}{dt}(x-\xi)^{\mathrm{T}} f'((1-t)\xi + tx)^{\mathrm{T}}$$

$$= (x-\xi)^{\mathrm{T}} f''((1-t)\xi + tx)^{\mathrm{T}}(x-\xi).$$

Writing $\eta = (1-\tau)\xi + tx$, we obtain that $F''(\tau)^{\mathrm{T}} = (x-\xi)^{\mathrm{T}} f''(\eta)^{\mathrm{T}}(x-\xi)$. Transposing both sides then yields $F''(\tau) = (x-\xi)^{\mathrm{T}} f''(\eta)(x-\xi)$ as required.

If ξ is a stationary point, then $f'(\xi) = 0$ and so the preceding theorem reduces to

$$f(\xi + h) - f(\xi) = \tfrac{1}{2} h^{\mathrm{T}} f''(\eta) h$$

where η lies on the line segment between ξ and $\xi + h$. In order that ξ correspond to a local maximum it is therefore clearly sufficient that $h^{\mathrm{T}} f''(\eta) h$ be *negative* for all h of sufficiently small length. The analogous condition for a local minimum is that $h^{\mathrm{T}} f''(\eta) h$ be *positive* for all h of sufficiently small length. What properties of the matrix $f''(\eta)$ guarantee one or other of these criteria? To answer this question reasonably elegantly requires the use of rather more linear algebra than we have been assuming so far.

19.37 Eigenvalues

If M is an $n \times n$ *symmetric* matrix, there exists an orthogonal matrix H (i.e. a matrix H satisfying $H^{\mathrm{T}}H = I$) such that

$$H^{\mathrm{T}} M H = D$$

where D is a diagonal matrix, i.e.

$$D = \begin{pmatrix} \lambda_1 & 0 \ldots 0 \\ 0 & \lambda_2 \ldots 0 \\ \vdots & \\ 0 & 0 \ldots \lambda_n \end{pmatrix}$$

The entries $\lambda_1, \lambda_2, \ldots, \lambda_n$ on the diagonal of D are called the *eigenvalues* of M and can be found by solving the polynomial equation

$$\det(M - \lambda I) = 0$$

obtained by setting the determinant of the square matrix $M - \lambda I$ equal to zero. It is a useful fact that the eigenvalues of a symmetric matrix with real entries are always real numbers.

We shall take this theory for granted. Its interest in the current context is that it allows us to discuss quadratic forms in a coherent fashion. If M is an $n \times n$ symmetric matrix and x is an $n \times 1$ column vector, we call

$$x^{\mathrm{T}} M x$$

a *quadratic form*. If $x^{\mathrm{T}}Mx > 0$ for all $x \neq 0$ we say that M is *positive definite*. If $x^{\mathrm{T}}Mx < 0$ for all $x \neq 0$ we say that M is *negative definite*. In view of the remarks at the end of §19.34, these notions are clearly relevant to the classification of stationary points.

If we write $x = Hy$, then $x^{\mathrm{T}} = y^{\mathrm{T}}H^{\mathrm{T}}$ and so

$$\begin{aligned} x^{\mathrm{T}}Mx &= y^{\mathrm{T}}H^{\mathrm{T}}MHy = y^{\mathrm{T}}Dy \\ &= \lambda_1 y_1^2 + \lambda_2 y_2^2 + \ldots + \lambda_n y_n^2. \end{aligned}$$

Since $\|x\|^2 = x^{\mathrm{T}}x = y^{\mathrm{T}}H^{\mathrm{T}}Hy = y^{\mathrm{T}}Iy = y^{\mathrm{T}}y = \|y\|^2$, we have that $x = 0$ if and only if $y = 0$. It follows that M is positive definite if and only if all its eigenvalues are positive. Similarly M is negative definite if and only if all its eigenvalues are negative. We may also deduce the useful inequality

$$\lambda_1\|x\|^2 \leqslant x^{\mathrm{T}}Mx \leqslant \lambda_n\|x\|^2 \tag{1}$$

where λ_1 is the smallest and λ_n is the largest of the eigenvalues of M.

If all the eigenvalues of M are non-negative, then (1) shows that $x^{\mathrm{T}}Mx$ has a minimum when $x = 0$. Similarly, if all the eigenvalues of M are non-positive, then $x^{\mathrm{T}}Mx$ has a maximum when $x = 0$. If one or more of the eigenvalues are zero, however, there are straight lines through the point 0 on which $x^{\mathrm{T}}Mx$ is identically zero. To see this we need the idea of an eigenvector.

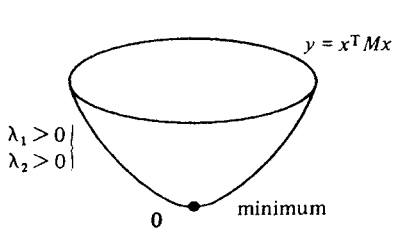
The unit vector $u^j = He^j$, where e^j is the column vector whose co-ordinates are all zero except for the jth which is equal to one, is an *eigenvector* corresponding to λ_j. If $x = tu^j$, then

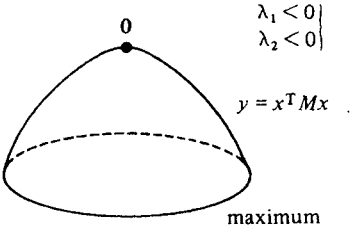
$$x^{\mathrm{T}}Mx = \lambda_j t^2. \tag{2}$$

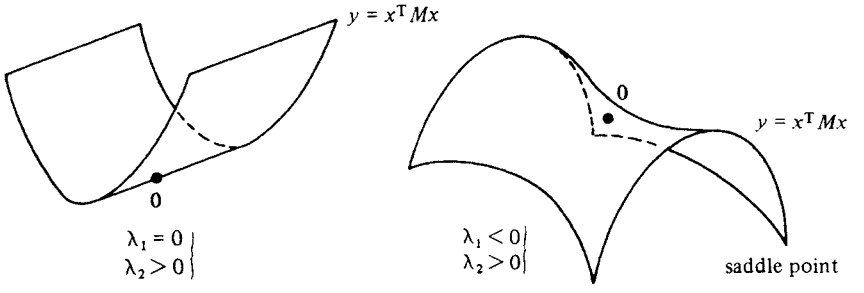
Thus, if $\lambda_j = 0$, then $x^{\mathrm{T}}Mx = 0$ for all x on the line through 0 which points in the direction of the eigenvector u^j.

It is also instructive to see what happens when some eigenvalues are positive and others are negative. Suppose in particular that $\lambda_1 < 0$ while $\lambda_n > 0$. From (2) it follows that $x^{\mathrm{T}}Mx < 0$ for $x \neq 0$ on the straight line through 0 in the direction of u^1. Similarly, $x^{\mathrm{T}}Mx > 0$ for $x \neq 0$ on the straight line through 0 in the direction of u^n. Thus, if we restrict our attention to values of x along one line through 0, we find that $x^{\mathrm{T}}Mx$ is strictly smallest when $x = 0$. But along a second line, $x^{\mathrm{T}}Mx$ is strictly largest when $x = 0$. Under these circumstances, we call 0 a *saddle point*.

The diagrams below illustrate the various cases we have considered.







19.38 Principal minors

If A and B are square $n \times n$ matrices, then $\det(AB) = (\det A)(\det B)$. It follows that

$$\begin{aligned}\det M &= \det(H^T DH) = (\det H^T)(\det D)(\det H) \\ &= \det(H^T H)\det D = \det I \det D.\end{aligned}$$

But the determinant of a diagonal matrix is simply the product of its diagonal entries. Hence

$$\det M = \lambda_1 \lambda_2 \ldots \lambda_n.$$

Various useful consequences follow. For example, M has a zero eigenvalue if and only if $\det M = 0$. Also $\det M$ is positive when M is positive definite and $\det M$ has the sign of $(-1)^n$ when M is negative definite. This last sentence is part of a more general result concerning the principal minors of M.

If

$$M = \begin{pmatrix} m_{11} & m_{12} & \ldots & m_{1n} \\ m_{21} & m_{22} & \ldots & m_{2n} \\ \vdots & & & \\ m_{n1} & m_{n2} & \ldots & m_{nn} \end{pmatrix}$$

then its *principal minors* are the determinants

$$P_1 = m_{11},\ P_2 = \begin{vmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{vmatrix},\ P_3 = \begin{vmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{vmatrix}$$

and so on, ending up with $P_n = \det M$. We then have the following results:

(I) The matrix M is positive definite if and only if

$$P_j > 0 \quad (j = 1, 2, \ldots, n).$$

(II) The matrix M is negative definite if and only if

$$(-1)^j P_j > 0 \quad (j = 1, 2, \ldots, n).$$

We shall take these results from linear algebra for granted. They are an extremely useful means for determining whether or not a matrix is positive or negative definite since actually to calculate the eigenvalues when $n \geqslant 3$ can be very difficult indeed. (Note incidentally that, if $P_j \geqslant 0$ $(j = 1, 2, \ldots, n)$ it does *not* follow that the eigenvalues of M are necessarily non-negative. The eigenvalues of

$$M = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

are $-1, 0$ and 2 but all the principal minors are non-negative. Nor is it true that $(-1)^j P_j \geqslant 0$ implies that the eigenvalues are necessarily non-positive.)

19.39 Examples

(i) The matrix

$$A = \begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix}$$

is positive definite because its principal minors are 5 and

$$\begin{vmatrix} 5 & -3 \\ -3 & 5 \end{vmatrix} = 25 - 9 = 16.$$

(ii) The matrix

$$B = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}$$

is negative definite because its principal minors are -2 and

$$\begin{vmatrix} -2 & 1 \\ 1 & -2 \end{vmatrix} = 4 - 1 = 3.$$

(iii) The matrix

$$C = \begin{pmatrix} 1 & 6 \\ 6 & 4 \end{pmatrix}$$

is neither positive definite nor negative definite because

$$\det C = \begin{vmatrix} 1 & 6 \\ 6 & 4 \end{vmatrix} = 4 - 36 = -32 < 0.$$

Note that, since $\det C \neq 0$, neither eigenvalue is zero.

19.40 Classification of stationary points

Suppose that f is a real-valued function with a stationary point at ξ. If all the second order partial derivatives of f exist and are continuous at each x satisfying $\|x-\xi\|<\Delta$, it is often possible to determine whether ξ corresponds to a local maximum, a local minimum or a saddle point of f by examining the matrix $f''(\xi)$. Since f has a stationary point at ξ, $f'(\xi)=0$ and so theorem 19.36 reduces to

$$f(\xi+h)-f(\xi) = \tfrac{1}{2}h^{\mathrm{T}}f''(\eta)h$$

provided $\|h\|$ is sufficiently small. We need to relate the quadratic form $h^{\mathrm{T}}f''(\eta)h$ to the quadratic form $h^{\mathrm{T}}f''(\xi)h$. For this purpose we use the inequality

$$|h^{\mathrm{T}}Lh| \leqslant \left\{\sum_{i=1}^{n}\sum_{j=1}^{n} l_{ij}^2\right\}^{1/2}\|h\|^2$$

which holds for any $n\times n$ matrix L and any $n\times 1$ column vector h. The proof consists of two applications of the Cauchy–Schwarz inequality. We have that

$$\begin{aligned}|h^{\mathrm{T}}Lh| &= \Big|\sum_i h_i \sum_j l_{ij}h_j\Big| \\ &\leqslant \sum_i |h_i|\left\{\sum_j l_{ij}^2\right\}^{1/2}\|h\| \\ &\leqslant \left\{\sum_i\sum_j l_{ij}^2\right\}^{1/2}\|h\|^2.\end{aligned}$$

19.41 Lemma Suppose that f is a real-valued function with second order partial derivatives which exist and are continuous at each $x\in\mathbb{R}^n$ satisfying $\|x-\xi\|<\Delta$. Then, given any $\epsilon>0$, we can find a $\delta>0$ such that

$$|h^{\mathrm{T}}(f''(\xi)-f''(\eta))h|<\epsilon\|h\|^2$$

provided that $0<\|h\|<\delta$ and η is as in theorem 19.36.

Proof We simply observe that

$$|h^{\mathrm{T}}(f''(\xi)-f''(\eta))h| \leqslant \left\{\sum_i\sum_j (f_{ij}(\xi)-f_{ij}(\eta))^2\right\}^{1/2}\|h\|^2$$

and then use the fact that each of the second order partial derivatives f_{ij} is continuous at ξ.

19.42 Theorem Suppose that f is a real-valued function with second order partial derivatives which exist and are continuous at each $x\in\mathbb{R}^n$ satisfying $\|x-\xi\|<\Delta$. Then

(i) If $f'(\xi)=0$ and $f''(\xi)$ is negative definite then f has a local maximum at ξ.

(ii) If $f'(\xi)=0$ and $f''(\xi)$ is positive definite, then f has a local minimum at ξ.

Proof We prove (ii). The proof of (i) is entirely analogous.

Let λ_1 be the smallest eigenvalue of $f''(\xi)$. It follows from (1) of §19.37 that

$$h^{\mathrm{T}} f''(\xi) h \geqslant \lambda_1 \|h\|^2$$

for all h.

Since $f''(\xi)$ is positive definite, $\lambda_1 > 0$ and so we may take $\epsilon = \frac{1}{2}\lambda_1$ in lemma 19.41. From theorem 19.36, we then obtain that

$$\begin{aligned} f(\xi + h) - f(\xi) &= \tfrac{1}{2} h^{\mathrm{T}} f''(\eta) h \\ &= \tfrac{1}{2} h^{\mathrm{T}} f''(\xi) h + \tfrac{1}{2} h^{\mathrm{T}} (f''(\eta) - f''(\xi)) h \\ &> \tfrac{1}{2}\lambda_1 \|h\|^2 - \tfrac{1}{2}\epsilon \|h\|^2 \end{aligned}$$

provided that $0 < \|h\| < \delta$. Hence, for $0 < \|h\| < \delta$,

$$f(\xi + h) - f(\xi) > \tfrac{1}{4}\lambda_1 \|h\|^2$$

and so f has a local minimum at ξ.

19.43 Theorem Suppose that f is a real-valued function with second order partial derivatives which exist and are continuous at each $x \in \mathbb{R}^n$ satisfying $\|x - \xi\| < \Delta$.

If $f'(\xi) = 0$ but $f''(\xi)$ is neither positive definite nor negative definite, then f has a (local) saddle point at ξ provided that

$$\det f(\xi) \neq 0.$$

Proof The conditions guarantee that the smallest eigenvalue λ_1 of $f''(\xi)$ is negative while the largest eigenvalue λ_n is positive. Using (2) of §19.37 together with the argument of theorem 19.42 we obtain that

$$f(\xi + h) - f(\xi) < \tfrac{1}{4}\lambda_1 \|h\|^2$$

for values of h which satisfy $0 < \|h\| < \delta$ and lie on the line through 0 in the direction u^1. Similarly

$$f(\xi + h) - f(\xi) > \tfrac{1}{4}\lambda_n \|h\|^2$$

for values of h which satisfy $0 < \|h\| < \delta$ and lie on the line through 0 in the direction u^n.

19.44 Note The condition $\det f(\xi) \neq 0$ in theorem 19.43 guarantees that no eigenvalues of $f''(\xi)$ are zero. It is worth pointing out that nothing very useful can be deduced about the behaviour of f at a stationary point ξ from the fact that an eigenvalue λ_j of $f''(\xi)$ is zero. In this case

$$f(\xi + h) - f(\xi) = \tfrac{1}{2} h^{\mathrm{T}} (f''(\eta) - f''(\xi)) h$$

for sufficiently small values of h on the line through 0 in the direction u^j and we have no information about the sign of the right hand side. (When $\lambda_j \neq 0$ this is irrelevant since the term is then dominated by $\frac{1}{2}\lambda_j \|h\|^2$.)

In the case $n = 1$, $f''(\xi)$ is just a real number and theorem 19.42 is the familiar result which relates the existence of a local maximum or minimum at a stationary point with the sign of the real number $f''(\xi)$. Theorem 19.43 does not apply in the case $n = 1$ because, if $f''(\xi)$ is neither positive or negative it must be zero.

In the case $n = 2$, it is best to begin examining $f''(\xi)$ at a stationary point ξ by looking at

$$\begin{aligned}\det f''(\xi) &= \begin{vmatrix} f_{11}(\xi) & f_{12}(\xi) \\ f_{21}(\xi) & f_{22}(\xi) \end{vmatrix} \\ &= f_{11}(\xi)f_{22}(\xi) - f_{12}(\xi)^2.\end{aligned}$$

We know from §19.38 that for $f''(\xi)$ to be positive definite or negative definite it is necessary that $\det f''(\xi) > 0$. If $\det f''(\xi) > 0$ and $f_{11}(\xi) > 0$, then $f''(\xi)$ is positive definite and we have a local minimum. If $\det f''(\xi) > 0$ and $f_{11}(\xi) < 0$, then $f''(\xi)$ is negative definite and we have a local maximum. If $\det f''(\xi) < 0$, theorem 19.43 assures the existence of a saddle point.

Do *not* be tempted to draw conclusions when $\det f''(\xi) = 0$. One of the two eigenvalues is then zero and, as explained in note 19.44, the method does not work in this case. It is not possible to be tempted to draw a wrong conclusion from $\det f''(\xi) > 0$ and $f_{11}(\xi) = 0$ because the latter equation implies that $\det f''(\xi) = -f_{12}(\xi)^2$.

19.45 Example Consider the function $f\colon \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x_1, x_2) = x_1^3 - 3x_1x_2^2 + x_2^4$. The stationary points are found by solving

$$\left.\begin{aligned} f_1(x_1, x_2) &= 3x_1^2 - 3x_2^2 = 0 \\ f_2(x_1, x_2) &= -6x_1x_2 + 4x_2^3 = 0 \end{aligned}\right\}$$

simultaneously. The solutions are $(3/2, 3/2)$, $(3/2, -3/2)$ and $(0, 0)$. The second derivative is

$$f''(x_1, x_2) = \begin{pmatrix} f_{11}(x_1, x_2) & f_{12}(x_1, x_2) \\ f_{21}(x_1, x_2) & f_{22}(x_1, x_2) \end{pmatrix} = \begin{pmatrix} 6x_1 & -6x_2 \\ -6x_2 & -6x_1 + 12x_2^2 \end{pmatrix}$$

and so the principal minors are

$$\det f''(x_1, x_2) = \begin{vmatrix} 6x_1 & -6x_2 \\ -6x_2 & -6x_1 + 12x_2^2 \end{vmatrix} = 36(-x_1^2 + 2x_1x_2^2 - x_2^2)$$

and

$$f_{11}(x_1, x_2) = 6x_1.$$

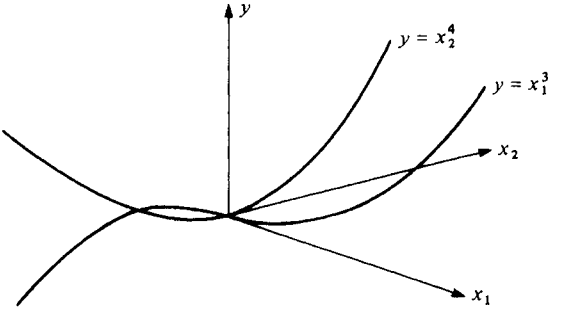
(i) Since $\det f''(3/2, 3/2) = 81 > 0$ and $f_{11}(3/2, 3/2) = 9 > 0$, we have that $(3/2, 3/2)$ is a local minimum.

(ii) Also $\det f''(3/2, -3/2) = 81 > 0$ and $f_{11}(3/2, -3/2) = 9 > 0$ and so $(3/2, -3/2)$ is also a local minimum.

(iii) Since $\det f''(0,0) = 0$, the method is useless for classifying the stationary point $(0, 0)$. Observe, however, that

$$y = f(0, x_2) = x_2^4$$
$$y = f(x_1, 0) = x_1^3$$

and so the function certainly does not have a local maximum or minimimum at $(0, 0)$.



19.46 Example Consider the function $f\colon \mathbb{R}^3 \to \mathbb{R}$ defined by $f(x_1, x_2, x_3) = x_1x_2 + x_2x_3 + x_3x_1$. The stationary points are found by solving the equations

$$\left.\begin{aligned} x_2 + x_3 &= 0 \\ x_1 + x_3 &= 0 \\ x_2 + x_1 &= 0 \end{aligned}\right\}$$

simultaneously. The only stationary point is therefore $(0, 0, 0)$. The second derivative at this point is

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

which has principal minors $2, -1$ and 0. The matrix is neither positive definite nor negative definite because one of the principal minors is zero. But $f''(0, 0, 0) \neq 0$ and so we have a saddle point at $(0, 0, 0)$.

19.47 Exercise

(1) A function $f\colon \mathbb{R}^n \to \mathbb{R}$ is defined by

$$f(x) = (Ax + a)^{\mathrm{T}}(Bx + b)$$

where A and B are $m \times n$ matrices and x, a and b are $m \times 1$ column

vectors. Find a system of linear equations satisfied by the local maxima and minima of f. [Hint: example 19.30.]

(2) Show that that the function $f\colon \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x_1, x_2) = x_1 e^{-x_2}$$

is twice differentiable at every point and determine $f''(x_1, x_2)$.

(3) Show that the function $f\colon \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x_1, x_2) = \begin{cases} \dfrac{x_1^3 x_2}{x_1^2 + x_2^2} & (x_1, x_2) \neq (0,0) \\ 0 & (x_1, x_2) = (0,0) \end{cases}$$

has a continuous first derivative $f'\colon \mathbb{R}^2 \to \mathbb{R}^2$ at $(0,0)$ and that all its second order partial derivatives exist at $(0,0)$. Prove that

$$f_{12}(0,0) \neq f_{21}(0,0)$$

and explain why this does not contradict proposition 19.33.

(4) Classify the stationary points of the function $f\colon \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x_1, x_2) = x_1^2 x_2 + x_2^3 x_1 - x_1 x_2$$

using the second derivative. [Hint: example 19.10.]

(5) Show that there exists no value of $\xi \in (0, 1)$ such that

$$f(1) - f(0) = f'(\xi)(1 - 0)$$

in the case of the function $f\colon \mathbb{R}^1 \to \mathbb{R}^2$ defined by

$$f(t) = \begin{pmatrix} t^2 \\ t^3 \end{pmatrix}.$$

(See §19.34.)

(6) Suppose that λ is an eigenvalue of an $n \times n$ symmetric matrix M with real entries. Using the results quoted in this chapter, prove that

$$\lambda^2 \leqslant \sum_{i=1}^{n} \sum_{j=1}^{n} m_{ij}^2.$$

19.48† Differentials

The derivative $f'(x)$ or $Df(x)$ of a vector function f at a point x was defined to be a matrix. Recall that f is differentiable at x if and only if there exists an affine function g which is tangent to f at x. The affine function g is given by

$$g(x + h) = f(x) + f'(x)h.$$